Problems in Engineering Physics I

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1 | Units & 1D Kinematics

1. Conversion factors are just fancy ways of writing 1. For example, 1 inch is the same distance as 2.54 cm. When two things are the same in math, we say they are equal and can write that as

$$1 \text{ inch} = 2.54 \text{ cm}. \tag{1.1}$$

This can be rearranged algebraically so that one side of the equation is 1, and the other side is the ratio of the two numbers

$$1 = \frac{2.54 \text{ cm}}{1 \text{ inch}}, \quad \text{or} \quad \frac{1 \text{ inch}}{2.54 \text{ cm}} = 1. \tag{1.2}$$

Since we can multiply anything by 1 without changing it, conversion factors let us go between different systems of units so that the quantities are still equal to each other. Using this knowledge, complete the following steps:

   (a) Using Google (or your search engine of choice), find how many feet are in a mile, how many kilometers in a mile, how many meters in a parsec, how many seconds in an hour, and how many Fahrenheit degrees (F°) in a Celsius degree (C°)[1]. Write these two things as equal to each other to get the ratios as shown in equation (1.1).

   (b) A marathon is 26.2 miles. Convert this distance to feet. (*Ans: 138,336 feet*)

   (c) Convert the distance in part (b) to centimeters (cm). (*Ans: 4,216,481.28 cm* [2])

   (d) Han Solo famously completed the Kessel Run in 12 parsecs (a unit of distance used in astronomy). What is this distance in feet? (*Ans:* 1.215×10^{18} *feet*)

   (e) On a particular day, the outdoor temperature starts at 28°F and gets up to 64°F. By how many Celsius degrees did the temperature change? (*Ans: 20 C°*)

2. For the numbers below, if it is given in long format, write it in powers of ten notation, if it is given in powers of ten notation, write it in long format. When going from long format to powers of ten, you may round to 3 significant figures.

   (a) 299,792,458

   (b) 4.21×10^{-3}

   (c) 0.000008675309

[1] This last one might be tricky, but you can figure it out if you remember that there are 100 C° between the freezing point and boiling point of water, while there are 180 F° between these same points.

[2] If your answer here is different in the last few digits, it doesn't mean you have made a mistake. You may simply have used an approximate conversion to go between imperial units (feet, inches, etc) and metric or SI units. The exact conversion is given by 1 inch = 2.54 cm.

(d) 1.36×10^9

3. You are walking to Engineering Physics one morning from your dorm just down the road, 500 m away. You make it 4/5 of the way there, before you realize that you left your homework in your dorm room. You run back, grab it, and then run to class. Assume all this motion is in a straight line.

   (a) What is the distance you traveled? (*Ans: 1300 m*)

   (b) What is your displacement? (*Ans: 500 m*)

4. An Olympic sprinter is training at a track. They run 100 m down the track in 11 s. They then slow down and come to rest over the next 10 m in 2.0 s, after which they turn around and walk back to the 110 m back to the starting line over the next 98 s. Assume that all the motion is in a straight line and take the direction of the sprint to be the positive direction.

   (a) What is the average speed of the sprinter while running the 100 m down the track? What is their displacement? What is their average velocity during this part of the motion? (*Ans:* $\overline{v} = 9.09$ *m/s,* $\vec{\mathbf{v}} = +9.09$ *m/s*)

   (b) What is the average speed of the sprinter if you include the slowing down after the sprint? What is their displacement? What is their average velocity during this part of the motion? (*Ans:* $\overline{v} = 8.46$ *m/s,* $\vec{\mathbf{v}} = +8.46$ *m/s*)

   (c) What is the average speed of the sprinter over the full motion, sprinting, slowing down, and walking back? What is their displacement? What is their average velocity here? Why is that their average velocity? (*Ans:* $\overline{v} = 1.98$ *m/s,* $\vec{\mathbf{v}} = 0$)

5. Let's examine the acceleration and deceleration of the sprinter in the previous problem. Assume that the sprinter reaches their top speed in the first 18 m of the sprint.

   (a) If the sprinter is accelerating for 18 m and then running at a constant speed for the remaining 82 m, what is the sprinter's acceleration during the first 18 m? **Hint:** You can't just use the average speed you found in the previous problem, but you can use the constant acceleration kinematics equations and some clever thinking. (*Ans: 3.20* m/s^2)

   (b) What is the sprinter's top speed? (*Ans: 10.73 m/s*)

   (c) If they slow down from their top speed as described in the previous problem, what is their average acceleration during that time? (*Ans: -5.365* m/s^2)

6. A rocket is in an isolated part of space, i.e. gravitational effects are negligible. At a point in time, it fires its engine. Taking the time it starts to fire it's engine as $t = 0$, it's motion during the burn is described by the equation

$$x(t) = At + Bt^2 + Ct^3, \tag{1.3}$$

   where x is in meters, t is in seconds, and A, B, and C are constants.

   (a) What are the units of the constants A, B, and C? What does that suggest the constants physically represent?

   (b) What would be the equation that describes the velocity as a function of time?

   (c) What would be the equation that describes the acceleration as a function of time?

(d) If the rocket starts with a velocity of 1462 m/s and ends with a velocity of 2166 m/s 180 s later, traveling 32,700 m in the process, what are the numerical values of A, B, and C? (*Ans: A = 1462, B = -25.25, C = 0.1076, units intentionally omitted due to part (a)*)

7. The acceleration of a race car is given by the function

$$a(t) = A\cos(t/10)\sin^2(t) = \frac{A}{2}\left[\cos\left(\frac{t}{10}\right) - \frac{1}{2}\cos\left(-\frac{19}{10}t\right) - \frac{1}{2}\cos\left(\frac{21}{10}t\right)\right] \tag{1.4}$$

where A is in units of m/s^2, and t is in seconds. The second equality comes from the magic of trigonometric identities, namely

$$\sin^2\theta = \frac{1-\cos(2\theta)}{2}, \quad \text{and} \quad 2\cos\theta\cos\phi = \cos(\theta-\phi) + \cos(\theta+\phi). \tag{1.5}$$

The car completes the quarter-mile drag in 15.708 s ($\sim 5\pi$ s).

(a) What is the expression for the car's velocity as a function of time?

(b) What is the expression for the car's position as a function of time?

(c) If $A = 8.04$ m/s^2, create individual graphs of the acceleration, velocity, and position as functions of time.[3]

8. You want to perform an experiment as a test of free-fall. You find a pool that has a 10 m high dive platform and the agree to allow you to come in after-hours to drop a ball and time how long it takes to hit the water. Your plan is to start your timer as you drop the ball, and stop it when you hear the splash of the ball in the water.

(a) Using your kinematics knowledge, how long should it take the ball to hit the water? (*Ans: 1.429 s*)

(b) As we will learn later on, sound travels at a finite speed of 343 m/s. How long will it take the sound of the splash to reach you? (*Ans: 0.0292 s*)

(c) What time do you expect your timer to read? (*Ans: 1.458 s*)

(d) What would your percent error in g be if you didn't account for the time for the sound to reach your ears? (*Ans: 3.96%*)

9. While hiking through the woods in a mountainous region, you find a cliff above a lake. Curious about the distance to down to the water's surface, you pick up a rock and drop it over the edge. You hear the splash of the rock 4.52 s after you dropped it. How high above the water is the cliff's edge? (*Ans: 89.0 m*)

[3] You can do this in any spreadsheet software, e.g. Microsoft Excel, Google Sheets, or LibreOffice Calc. Start by labeling the first column as t setting the first value below the label to 0, then in the cell below the zero enter "=A2 + 0.05*pi()". Next, mouse over the bottom right corner of the cell you entered this equation in, until the cursor changes to a + sign, click and drag down to fill in values for time. Do this until you reach 5π s. Label the next three columns as a, v, and x, respectively, and use your equations to compute the values. Pro-tip, enter the equations in the cells next to $t = 0$, then highlight those three cells, mouse over the bottom-right corner of the right most cell until your cursor becomes a + sign, and double-click. The cells below will autofill for each value of time. Lastly, highlight the time column and the acceleration column and insert a "chart", i.e. a plot or graph, use the scatter plot with smooth lines. Repeat that for the other two plots.

10. **Numerical:** Often times in physics, we end up with equations that are very difficult to solve exactly. In those cases, we can use numerical methods to help us get an approximate answer. For example, we often will ignore air resistance in introductory level physics classes because the acceleration it causes has the form of

$$a_{\mathrm{D}} = -\alpha v^2, \tag{1.6}$$

where α is a constant that depends on the objects shape and mass, and the minus sign reflects that it will oppose the direction of motion. Since this acceleration is now dependent on the velocity, it becomes a much more difficult problem to solve analytically.

Consider dropping a ball with $\alpha = 8.90 \times 10^{-3}\ \mathrm{m}^{-1}$ from a great height. We will take down to be the positive y-direction so we can simply count how far the ball has fallen.

(a) Using a time step of $\Delta t = 1 \times 10^{-4}$ s, calculate the position, velocity, and acceleration of the ball both with and without drag. Note, the accelerations will be

$$a = (9.8 - \alpha v^2)\ \mathrm{m/s}^2 \tag{1.7}$$

with drag, and

$$a = 9.8\ \mathrm{m/s}^2 \tag{1.8}$$

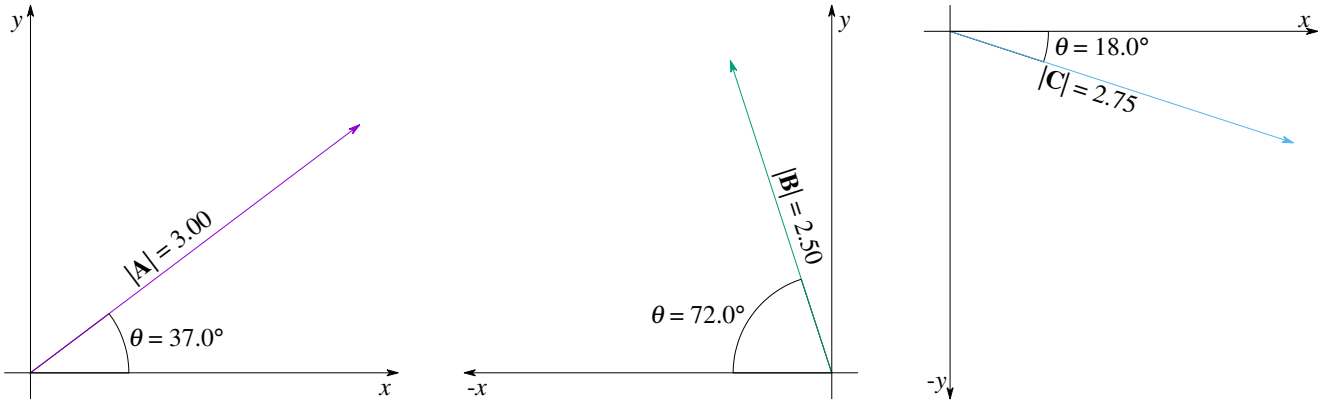
without drag. See the "Solving Numerical Problems" section of this book for details on how to do this.

(b) Create a graph comparing the distance fallen over time in these two cases. Looking at this graph, at about what time does the distance fallen for the two cases start to diverge?

(c) Create a graph comparing the velocity of the ball over time in these two cases. Looking at this graph, at about what time does the velocity for the two cases start to diverge? What is happening to the velocity of the ball with drag?

2 | Vectors & Kinematics

1. Consider the following three vectors, $\vec{\mathbf{A}}$, $\vec{\mathbf{B}}$, and $\vec{\mathbf{C}}$



   (a) Using a ruler and protractor, find the vector sum $\vec{\mathbf{A}} + \vec{\mathbf{B}} + \vec{\mathbf{C}}$ through graphical vector addition. To do this, define a convenient scale such as 1 unit = 3 cm, so that you can draw the vectors to scale. Once you have completed the graphical addition, measure the length and angle of the resultant, making sure to scale the length back to "units" from centimeters.

   (b) Break each vector into its x and y components and write them down using unit vector notation. (*Ans:* $\vec{\mathbf{A}} = 2.396\,\hat{\mathbf{i}} + 1.805\,\hat{\mathrm{j}}$, $\vec{\mathbf{B}} = -0.7725\,\hat{\mathbf{i}} + 2.378\,\hat{\mathrm{j}}$, $\vec{\mathbf{C}} = 2.615\,\hat{\mathbf{i}} - 0.8498\,\hat{\mathrm{j}}$)

   (c) Find the vector sum $\vec{\mathbf{A}} + \vec{\mathbf{B}} + \vec{\mathbf{C}}$ using component addition, then report the resultant as a magnitude and angle. (*Ans:* 5.392 at 38.18°)

   (d) Compare your result for the graphical vector addition with your result from the component addition. Do your results agree? Should they? If they disagree, which method gives the more accurate result? Why?

2. Consider the vectors

$$\vec{\mathbf{A}} = -2.36\,\hat{\mathbf{i}} + 2.99\,\hat{\mathbf{j}}, \quad \text{and} \quad \vec{\mathbf{B}} = 4.84\,\hat{\mathbf{i}} + 2.40\,\hat{\mathbf{j}} \tag{2.1}$$

   (a) Write $\vec{\mathbf{A}}$ in terms of its magnitude and direction. (*Ans:* 3.809 at 51.72° above -x-axis)

   (b) Write $\vec{\mathbf{B}}$ in terms of its magnitude and direction. (*Ans:* 5.402 at 26.375° above +x-axis)

   (c) Write $\vec{\mathbf{A}} + \vec{\mathbf{B}}$ in terms of its magnitude and direction. (*Ans:* 5.933 at 65.292° above +x-axis)

   (d) Write $\vec{\mathbf{B}} - \vec{\mathbf{A}}$ in terms of its magnitude and direction. (*Ans:* 7.224 at 4.6846° below +x-axis)

3. After your physics lab lets out, you want to walk over to the Walmart Neighborhood Market from Yates Hall. You start by walking south on Joplin Street for 318 m. Then you head east along Ford street for 530 m, followed by heading south of Homer Street for 64 m. This gets you to Normal Street which runs at an angle of 45° south of east. You follow Normal Street for 315 m. Next you head south of Taylor Street for 280 m, then east on Ohio Street for 452 m, south on Rouse for 126 m, and lastly west across the parking lot for 76 m to get to the door.

   (a) What is the total *distance* that you walked? (*Ans:* 2161 m)

   (b) What is your *displacement*? (*Ans:* $\vec{\mathbf{r}} = (1128.74\,\hat{\mathbf{i}} - 1010.74\,\hat{\mathbf{j}})$ m)

   (c) If it takes you 30 minutes to complete this walk, what was your average *speed*? (*Ans:* 1.20 m/s)

   (d) What was your average *velocity*? (*Ans:* $\vec{\mathbf{v}} = (0.627\,\hat{\mathbf{i}} - 0.5615\,\hat{\mathbf{j}})$ m/s)

4. A particle starts at $(x, y) = (0, 0)$ at time $t = 0$ with a velocity of

$$\vec{\mathbf{v}}_0 = (5.61\,\hat{\mathbf{i}} - 1.22\,\hat{\mathbf{j}}) \text{ m/s}. \tag{2.2}$$

   The particle experiences an acceleration of

$$\vec{\mathbf{a}} = (-0.86\,\hat{\mathbf{i}} + 0.31\,\hat{\mathbf{j}}) \text{ m/s}^2. \tag{2.3}$$

   (a) How long until the particle comes to rest in the x-direction? (*Ans:* 6.523 s)

   (b) What are the position and velocity of the particle at that maximum x position? (*Ans:* $\vec{\mathbf{r}} = (18.298\,\hat{\mathbf{i}} - 1.363\,\hat{\mathbf{j}})$ m, $\vec{\mathbf{v}} = (0.802\,\hat{\mathbf{j}})$ m/s)

   (c) How long until the particle comes to rest in the y-direction? (*Ans:* 3.935 s)

   (d) What are the position and velocity of the particle at that maximum y position? (*Ans:* $\vec{\mathbf{r}} = (15.417\,\hat{\mathbf{i}} - 2.401\,\hat{\mathbf{j}})$ m, $\vec{\mathbf{v}} = (2.226\,\hat{\mathbf{i}})$ m/s)

5. The position of a particle as a function of time is described by the vector equation

$$\vec{\mathbf{r}} = (7.0t^3 + 5.3t^4) \text{ m}\,\hat{\mathbf{i}} + (3.9t^6 - 3.9) \text{ m}\,\hat{\mathbf{j}}. \tag{2.4}$$

   (a) Find an expression for the particles velocity as a function of time.

   (b) Find an expression for the particles acceleration as a function of time.

   (c) At $t = 3.0$ s, what is the particles *displacement* from $t = 0$? (*Ans:* $\vec{\mathbf{r}} = (618.3\,\hat{\mathbf{i}} + 2843\,\hat{\mathbf{j}})$ m)

   (d) What is the particles average velocity over this time interval? (*Ans:* $\vec{\mathbf{v}}_{\text{avg}} = (206.1\,\hat{\mathbf{i}} + 947.7\,\hat{\mathbf{j}})$ m/s)

6. A baseball leaves the bat after being with a velocity of 54.0 m/s at 32.0° above the horizontal, starting at a height of about 0.80 m above the ground.

   (a) How long will it take for the baseball to hit the ground? (*Ans:* 5.868 s)

   (b) What will its velocity (magnitude and direction) be when it hits the ground? (*Ans:* 54.145 m/s at 32.245° below $+x$-axis)

   (c) What is the maximum height of the baseball? (*Ans:* 42.578 m)

   (d) If the baseball is heading straight for the center field fence, 125 m away, how long will it take to reach the fence? (*Ans:* 2.730 s)

(e) If the fence is 2.44 m tall, will this hit be a home run?

7. You are a stunt coordinator for a major action film. In a particular scene, the protagonist is supposed to jump from one cliff to another over a ravine on a motorcycle to escape from his pursuers. You build a ramp on each side of the ravine. Because the far-side cliff is a bit lower, the stuntman will land the motorcycle 5.0 m below where they take off. The ramp they will launch from is angled at 10.0° above the horizontal and they will able to get up to about 100 mph (45.0 m/s) before launching off the ramp. Assume that air resistance is negligible.

   (a) What is the time of flight? (*Ans:* 2.084 s)

   (b) You want the angle of the ramp on the far side to match the angle of the velocity of the stuntman and motorcycle when they land to make the landing as soft as possible. What should be the angle of the landing ramp? (*Ans:* 15.886°)

   (c) What is the horizontal distance between the take-off and landing points? (*Ans:* 92.368 m)

8. An archer is getting in some target practice. Their arrows leave the bow at 98.0 m/s, from a height of 1.7 m. The target is 70.0 m away with the bullseye 1.7 m above the ground. At what angle should the archer aim their arrow to hit the bullseye? (*Ans:* 2.05° above $+x$-axis)

9. You're a contestant on a wacky obstacle course game show. You have one remaining obstacle to clear in the final round, a conveyor belt that runs at 1.0 m/s in a direction perpendicular to the distance you need to traverse. You must run across this 5.0 m long contraption in 1.2 s in order to win. How fast and at what angle should you run in order to win? (*Ans:* 4.285 m/s at 13.496° below $+x$-axis if belt moves in $+y$-direction)

10. **Numerical:** We saw in the last homework that things get more complicated when we include the effects of air resistance. Here we have looked at projectile motion, so let's now figure out what happens when we include the effects of air resistance. Note, the acceleration of the projectile will be

$$\vec{\mathbf{a}} = \vec{\mathbf{g}} - \alpha v^2 \, \hat{\mathbf{v}} \tag{2.5}$$

where $\hat{\mathbf{v}}$ is the unit vector in the direction of the velocity. We can break this into components as

$$a_x = -\alpha v v_x, \quad \text{and} \quad a_y = -g - \alpha v v_y \tag{2.6}$$

where $v = \sqrt{v_x^2 + v_y^2}$ and we have assumed that down is the negative y-direction. We will take $alpha = 8.90 \times 10^{-3}$ m^{-1}.

   (a) Set up a spreadsheet or a program that will compute the position, velocity, and acceleration of the projectile in time steps of $\Delta t = 1 \times 10^{-4}$ s. Assume a launch velocity of 30.0 m/s at 25.0°. Have these values input in a way that they are simple to change, e.g. in a spreadsheet place them in cells near the top and then use fixed references to those cells where needed. Make sure to find the point where the ball hits the ground (i.e. y goes from positive to negative).

   (b) Write down the range (displacement in the x-direction), then start incrementing the launch angle (say by 5°), recording the range each time. You may need to extend the time of the 'simulation" in some cases for the projectile to reach the ground. When the range starts to decrease compared to the previous trial, go back to the last angle, then step through angles in smaller increments between these to hone in on the angle that maximizes the range.

3 | Newton's Laws & Forces

1. A train is heading towards a crossing where a car has stalled at 25.0 m/s. The train starts to apply the brakes 300 m away from the tracks. The train has a mass of 4.54×10^6 kg.

   (a) Draw and label a free-body diagram for the train as it is applying the brakes.

   (b) If the train is to stop in time, what is the needed acceleration? Use your kinematics knowledge here. (*Ans*: -1.042 m/s^2)

   (c) What is the needed force to bring the train to rest in time? (*Ans*: -4.729×10^6 N)

   (d) How does this force compare to the weight of the train? Calculate the ratio of the needed force to the weight of the train. (*Ans*: 10.6% of the weight)

2. You are sitting down to do some studying with your stack of 5 perfectly reasonably priced textbooks. Your physics textbook has a mass of 1.143 kg, your calculus book has a mass of 1.024 kg, your chemistry book has a mass of 1.234 kg, the novel for your literature course has a mass of 0.227 kg, and your programming book has a mass of 0.943 kg.

   (a) Assuming the books are stacked in the order that they are listed above, starting with the physics book sitting on your desk, draw and label a free-body diagram for each textbook.

   (b) Identify and list all the third law force pairs.

   (c) Write out the sum of forces equation for each book.

   (d) Calculate the weight of each book. (*Ans:* In order listed in the problem statement, 11.20 N, 10.04 N, 12.09 N, 2.22 N, 9.241 N)

   (e) Calculate the normal force acting on each book. (*Ans:* In order listed in the problem statement, 44.786 N, 33.586 N, 23.551 N, 11.46 N, 9.241 N)

3. At a fancy ice bar, there are three decorative metal blocks resting on top of the very smooth, icy surface of the bar, butted right up against each other. You apply a small force of 1.25 N to the left most block. The left most block has a mass of 1.00 kg, the middle block has a mass of 2.00 kg, and the last block has a mass of 0.75 kg.

   (a) Draw and label a free-body diagram for each block. Specifically list all third law force pairs.

   (b) What is the acceleration of the *system* of blocks? What does this mean about the acceleration of each individual block? (*Ans*: 0.333 m/s^2)

   (c) What is the magnitude of the force from the left most block on the middle block? (*Ans*: 0.917 N)

   (d) What is the magnitude of the force from the middle block on the right most block? (*Ans*: 0.250 N)

4. While visiting with your family for Christmas, you wake up on Christmas morning to find a nice layer of freshly fallen snow. Your young cousin asks you to take them out sledding. You find a hill that slopes at about 10° and is about 5 m long (this is the distance along the sloped surface).

   (a) Draw and label a free-body diagram of your cousin sitting in the sled while sliding down the hill. Assume that the sled slides on the snow with negligible friction.

   (b) If your cousin has a mass of 23 kg, and the plastic sled has a mass of 2 kg, what is the acceleration of the system consisting of your cousin and the sled? (*Ans*: 1.702 m/s^2)

   (c) How long will it take them to slide the full 5 m down the hill if they start from rest? (*Ans*: 2.424 s)

   (d) What is their speed at the bottom of the hill? (*Ans*: 4.126 m/s)

5. A factory worker is using a pulley to lift a crate. A rope is tied around the crate, goes through a pulley attached to the ceiling, and hangs down the other side where the worker can pull on it. The 50 kg crate is at rest on the floor. The worker applies a 525 N force to the rope by pulling down as hard as they can.

   (a) Draw and label a free-body diagram for the crate while it is being lifted up by the rope.

   (b) What is the acceleration of the crate? (*Ans*: a = 0.70 m/s^2)

   (c) Draw and label a free-body diagram for the pulley.

   (d) What is the tension in the support for the pulley from the ceiling? (*Ans*: 1050 N)

6. One of the fastest elevators in the world is found in Shanghai Tower, which reaches speeds of 20.5 m/s (about 46 mph). Assume that the mass of the elevator car is 1200 kg and that it accelerates at 2.45 m/s^2, with a 75 kg person standing inside.

   (a) Draw and label a free-body diagram for the elevator, as well as one for the person standing inside while it is accelerating upwards.

   (b) What is the tension in the cable that would support the elevator? (*Ans*: 1.562×10^4 N)

   (c) How long would it take the elevator to reach its top-speed? (*Ans*: 8.367 s)

   (d) What is the normal force on the person? (*Ans*: 918.75 N)

   (e) Draw and label free-body diagrams for the elevator and the person as the elevator is accelerating downwards.

   (f) What are the tension in the cable and the normal force on the person now? (*Ans*: $T = 9371.25$ N, $F_N = 551.25$ N)

7. Block A ($m_A = 8.48$ kg) rests on a frictionless plane inclined at 34.0°. A rope is attached to the block, run over a pulley at the top and attached to block B ($m_B = 4.74$ kg) which hangs freely.

   (a) Draw and label a free-body diagram for each block.

   (b) Write out the sum of forces equations for each block.

   (c) Find the magnitude and direction of the acceleration of the blocks. (*Ans*: $a = 1.450 \times 10^{-3}$ m/s^2 with A moving down the ramp, B moving upwards.)

   (d) What is the tension in the rope? (*Ans*: 46.459 N)

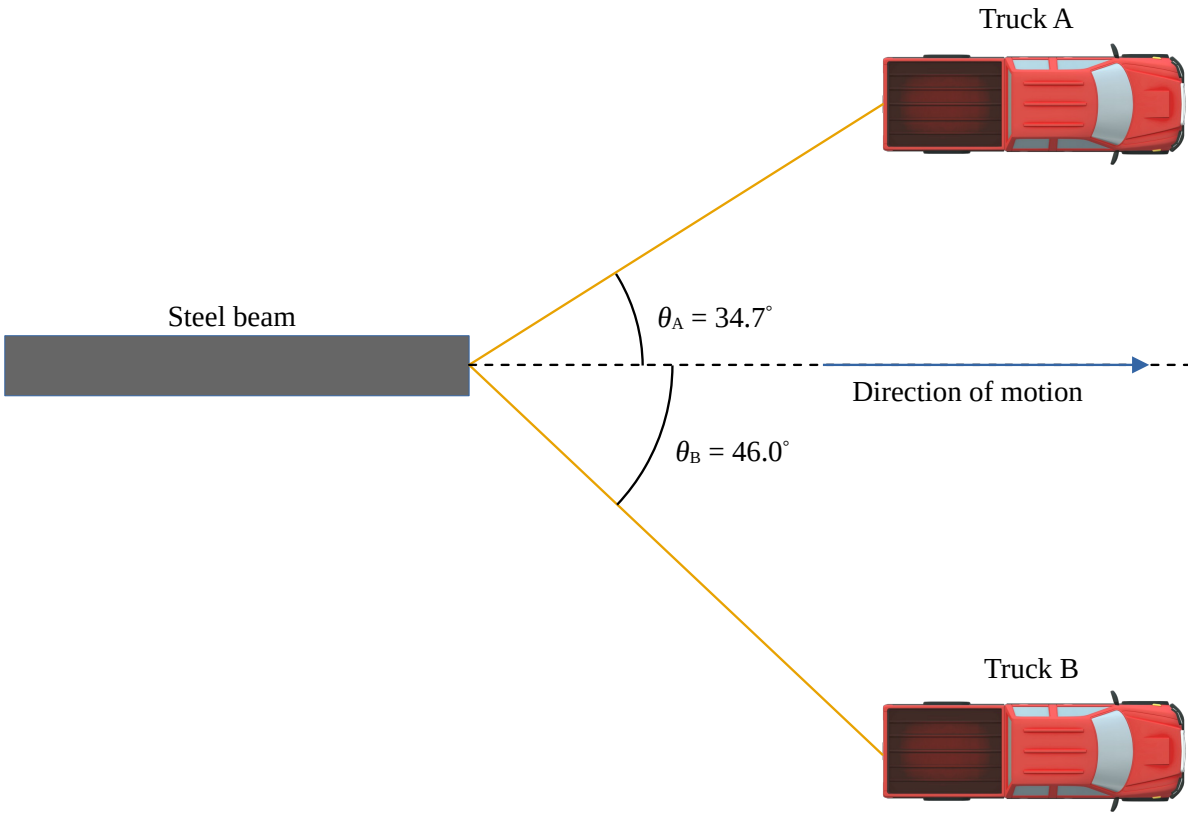
8. At a construction site, two pick-up trucks have tied ropes to a large steel I-beam. The beam is moving straight along the line as shown in the diagram below. The rope going to truck A is pulling on the beam at 34.7° with respect to the line the beam is moving along with a force 1627 N. The rope going to truck B is pulling on the beam at 46.0° with respect to the line the beam is moving along.

   (a) Draw and label a free-body diagram for the beam. Assume there is a friction force directed exactly opposite to the motion of the beam.

   (b) What must the net force perpendicular to the direction of the beam's movement be?

   (c) With this knowledge, what is the *magnitude* of the tension in the rope attached to truck B? (*Ans*: 1287.59 N)

   (d) If the beam is moving at a constant speed, what is the magnitude of the friction force acting on the beam? (*Ans*: 2232.06 N)



9. A child is pulling a 12 kg wagon down the sidewalk. They are pulling on the handle at an angle of 18° with respect to the horizontal with a magnitude of 35 N.

   (a) Draw and label a free-body diagram for the wagon. Assume that there is some net drag force (rolling friction, friction in the bearings, etc) acting on the wagon pointing in the opposite direction of the wagon's motion.

   (b) What is the weight of the wagon? (*Ans*: 117.6 N)

   (c) What is the normal force from the sidewalk on the wagon? (*Ans*: 106.78 N)

   (d) If the wagon is being pulled at a constant speed, what must the net drag force be? (*Ans*: 33.287 N)

10. **Numerical:** In a physics lab you are given a force table, a small circular table with angles marked in one degree incriments around its edge, and three pulleys that can be placed at any angle. There is a ring with three strings attached that allow you to hang weights off them to apply three forces to the ring. Your task is to balance the force table so that the net force acting on the ring is zero.

    (a) Draw a free-body diagram for the ring with one string (A) at 0°, i.e. in the x-direction and a second string (B) at an arbitrary angle between 1° and 179°. I'd recommend not choosing 90° so that you can get a clearer idea of how the forces will add. Picture the third string (C) at some other angle that should roughly balance the forces.

    (b) Using your free-body diagram as a guide, work out the equation that will give you the x and y components of the force from the third string given the x and y components of the other two forces.

    (c) Set up a spreadsheet with places to enter the massess hanging from the first and second strings (A and B), a column for the angles between 1° and 179°, a column that computes the x and y components of the force from the third string, a column that computes the needed mass (related to the magnitude of the third force), and a column that computes the angle of the third force[1].

    (d) Using this spreadsheet, compute the needed mass and angle for the third force if $m_A = 0.100$ kg, and $m_B = 0.150$ kg.

[1] Use the `atan2(x, y)` function for this, where you point to the x component first and then the y component. The normal `atan` function only returns angles between $-90°$ and 90°, while `atan2` will return angles between 0° and 180° for the first and second quadrants, and then $-180°$ and 0° for the third and fourth quandrants.

4 | Friction & Circular Motion

1. At a construction site, two pick-up trucks have tied ropes to a large steel I-beam ($m_{\text{beam}} = 1300$ kg). The beam is moving straight along the line as shown in the diagram below. The rope going to truck A is pulling on the beam at 34.7° with respect to the line the beam is moving along with a force 1627 N. The rope going to truck B is pulling on the beam at 46.0° with respect to the line the beam is moving along.

   (a) Draw and label a free-body diagram of the beam. Assume there is a force directed exactly opposite to the motion of the beam.

   (b) Assuming that the beam is moving at a constant speed, what is the magnitude of the friction force acting on the beam? (*Ans*: 2232.06 N)

   (c) What type of friction is acting on the beam, static or kinetic? How do you know?

   (d) What is the coefficient of friction between the beam and the ground? (*Ans:* 0.1752)

2. In lab you are asked to find the coefficients of friction between a block with a rubber pad glued onto the bottom and an aluminum inclinable plane that can be set to any angle you want.

   (a) Draw and label a free-body diagram for the block on the ramp inclined to an arbitrary angle θ.

   (b) Assuming that at this angle θ the block is just able to stay at rest, derive an expression for the coefficient of static friction.

   (c) You perform the experiment by placing the block on the inclinable plane and slowly changing the angle. As soon as the block starts to slide you stop and record the current angle. You perform 5 trials and measure the following angles: 24.7°, 25.1°, 25.2°, 25.3°, and 25.3°. What is the coefficient of static friction between the aluminum and the rubber? (*Ans*: 0.4689)

   (d) Now you want to find the coefficient of kinetic friction. You set the incline to 30° to be sure that the block will slide. You take a video of the block, put it into Tracker and determine that the acceleration of the block is 2.35 m/s^2. What is the coefficient of kinetic friction? (*Ans*: 0.3005)

3. Block A ($m_{\text{A}} = 8.48$ kg) rests on a plane inclined at 34.0°. A rope is attached to the block, run over a pulley at the top and attached to block B ($m_{\text{B}} = 7.74$ kg) which hangs freely.

   (a) Draw and label free-body diagrams for both block A and B.

   (b) What is the minimum coefficient of static friction between block A and the plane that would allow the system to remain at rest? (*Ans*: 0.4264)

   (c) What coefficient of kinetic friction would allow the blocks to slide at a constant speed? State your reasoning. (*Ans*: 0.4264)

4. Over winter recess, you and your friends head out to Colorado to do some skiing. You are at the top of the Pallavicini trail in Arapahoe Basin. This trail is angled at 32° and has a length along the slope of 2152 feet (655.9 m). The coefficient of kinetic between your waxed skis and the snow is 0.1.

   (a) Assuming that you go in a straight line down the slope, what is your acceleration? (*Ans*: 4.362 m/s^2)

   (b) If you start from rest at the top of the slope, how fast will you be going at the bottom? (*Ans*: 75.644 m/s)

   (c) If there is a long flat area at the bottom of the slope, and the coefficient of kinetic friction is still the same, how far would you travel before coming to a stop? (*Ans:* 2920.94 m)

   (d) How long would it take you to come to a stop? (*Ans*: 77.188 s)

5. While bored, you are swinging your keys ($m = 230$ grams) that are at the end of a 34 inch (86.36 cm) lanyard in a vertical circle.

   (a) Draw and label a free-body diagram for the keys at the top of the circular path.

   (b) What is the minimum speed that you can swing the keys so that they continue to move in a circle at the top of the arc? **Hint:** What is the lowest the acceleration can be at the top? (*Ans:* 2.909 m/s)

   (c) Draw and label a free-body diagram for the keys at the bottom of the circular path.

   (d) Assuming that you are adjusting the tension in the lanyard so that it continues to move at this same speed as it goes around the circle, what would the tension be at the bottom of the circlular path? (*Ans*:4.508 N)

6. You are working for the automotive company Bugatti, known for making some of the fastest cars (rich) people can buy. Your first assignment is working with a team to develop an oval test track that can allow Bugatti's cars to run, safely, at their top speeds of 250 mph (111.76 m/s) around the track. The curved ends of the oval will have a radius of curvature of 1500 m.

   (a) Draw and label a free-body diagram of a car on the banked, curved, section of the track assuming that no friction force is needed to keep the car on the track. For now, just use an arbitrary banking angle θ.

   (b) Write out the sum of forces equations and derive an equation for the banking angle of the track.

   (c) Using this equation, determine the banking angle needed for the speed given above. (*Ans*: 40.35°)

7. In order for humans to be able to spend extended times in space, it is going to be necessary to create artificial gravity on space stations and space craft. Being in zero or low gravity environments cause humans to quickly lose bone and muscle mass, which can be easily combated if there is something to mimic gravity. Let's say that we design a spaceship to be a giant ring with the engine in the very center of the ring and connected by supporting structure. The ring has a radius of 600 m. How many revolutions per minute would be needed to simulate Earth's surface gravity (9.8 m/s^2)? (*Ans*: 1.220 rpm)

8. A string is pinned to the center of a frictionless surface and attached to a block ($m_A = 50$ grams) that is moving in uniform circular motion in a circle with a radius of 13 cm. Another string runs from this block to another block ($m_B = 25$ grams) moving in a larger circle with radius of 19 cm. The system is completing 2 revolutions per second.

   (a) Draw and label free-body diagrams for both blocks.

   (b) What is the tension in the string that connects the two blocks to each other? (*Ans*: 0.75 N)

   (c) What is the tension in the inner string that connects the inner block to the middle? (*Ans*: 1.776 N)

9. Your designing a roller coaster and you want the riders to feel weightless as they go over the crest of the particular hill. The hill will have a radius of curvature of 10.0 m, and you assume that the average rider will have a mass of 70.0 kg. How fast should the train of cars be moving while going over the hill so that the riders will feel weightless? Does the mass of a particular rider affect your conclusion? (*Ans*: 9.899 m/s)

10. **Numerical:** When skydiving, you will reach a terminal velocity due to the velocity dependent drag force acting on you as you fall. As your velocity increases, the drag force increases until it exactly balances the gravitational force, at which point you simply fall at a constant speed.[1] The net force can be written as

$$\vec{\mathbf{F}} = m\frac{d\vec{\mathbf{v}}}{dt} = m\vec{\mathbf{g}} - kv^2\hat{\mathbf{v}}, \tag{4.1}$$

   where $k = 0.22$ kg/m, and the minus sign reflects the fact that the drag force will be in the opposite direction of the velocity of the skydiver. Taking up to be positive, we can see that the acceleration is given by

$$a = -g + \frac{k}{m}v^2 \tag{4.2}$$

   where all motion is in the y-direction.

   (a) Using a spreadsheet, compute the position, velocity, and acceleration as functions of time for a 70.0 kg skydiver for $0\text{ s} \leq t \leq 15.0\text{ s}$.

   (b) Create graphs for the position, velocity, and acceleration as functions of time.

   (c) Using your data, how long does it take to reach terminal velocity? Do your graphs support this conclusion? Make sure to write a discussion to support your conclusion.

[1] Actually, you will only asymptotically approach this terminal velocity, but never quite reach it as your acceleration will continually decrease as you get closer and closer.

5 | Gravity & Kepler's Laws

1. Consider masses located at the following coordinates

Particle	x (m)	y (m)	m (kg)
1	1.42	3.18	94.3
2	1.09	3.71	55.7
3	0.04	0.82	74.6
4	4.18	0.49	74.1
5	3.04	2.24	82.4

   What is the force of gravity, both magnitude and direction[1], of the gravitational force on mass 5? (*Ans*: 1.5185×10^{-7} N at 175.88° from $+x$-axis)

2. After you graduate, you get a job at NASA. One of your first assignments is working with a team that has sent a satellite to orbit and study a large asteroid. One of the reasons for using a satellite to study these asteroids is that we can accurately measure their masses by the speed at which the satellite needs to orbit.

   (a) Derive an expression for the mass of the asteroid in terms of the orbital radius, the orbital speed, and Newton's universal gravitational constant, G.

   (b) It is found that the satellite needs to maintain 377.8 m/s to maintain an orbit with a radius of 30 km. What is the mass of the asteroid? (*Ans*: 6.4198×10^{19} kg)

   (c) What is the period of the satellite's orbit? (*Ans*: 498.9 s)

3. The vast majority of the mass of our solar system is confined in the Sun, but how can we find the mass of the Sun? We know that the period of the Earth's orbit is about 365.25 days, and the average distance between Earth and the Sun is 1.47×10^{11} m. Using your knowledge of gravity and uniform circular motion, derive and expression for the mass of the Sun in terms of the radius and period of the Earth's orbit (along with some constants), then use that expression to estimate the mass of the Sun. (*Ans:* 1.89×10^{30} kg)

4. The Moon orbits the Earth, and the Earth-Moon system then orbits the Sun. Let's examine the gravitational forces on the Moon from the Earth and the Sun. The Moon has a mass of $M_{\text{M}} = 7.35 \times 10^{22}$ kg, the Earth has a mass of $M_{\text{E}} = 5.97 \times 10^{24}$ kg, and the Sun has a mass of $M_{\text{S}} = 1.99 \times 10^{30}$ kg. The average Earth-Moon distance is $r_{\text{EM}} = 3.84 \times 10^{8}$ m, and the average Earth-Sun distance (which is also basically the same as the average Moon-Sun distance) is $r_{\text{ES}} = 1.47 \times 10^{11}$ m.

[1] This can be given as the magnitude and an angle, or using unit vector notation.

(a) Calculate the gravitational force between the Moon and the Earth. (*Ans:* 1.98×10^{20} *N*)

(b) Calculate the gravitational force between the Moon and the Sun. (*Ans:* 4.51×10^{20} *N*)

5. You are standing on a scale while in an elevator. For a brief moment when the elevator starts moving you notice that the scale reads 1.2 times your usual weight. What is the acceleration of the elevator and are you going up or down? (*Ans:* 1.96 m/s^2, up)

6. After taking up amatuer astronomy, you decide to perform a little experiment to measure the mass of Jupiter. Four of Jupiter's moons are easily visible in a telescope and you decide to image them every night for two months. From your observations, you determine that the periods of the moons are

Moon	**Period** (days)	**Orbital Radius** (m)
Io	1.8	4.22×10^8
Europa	3.6	6.71×10^8
Ganymede	7.2	1.07×10^9
Callisto	16.7	1.88×10^9

(a) Obtain an estimate of the mass of Jupiter from the period of each moon. (*Ans:* In order of table 1.839×10^{27} kg, 1.848×10^{27} kg, 1.874×10^{27} kg, 1.889×10^{27} kg)

(b) Average the masses you found and calculate the standard deviation to get your best estimate of the mass of Jupiter. (*Ans*: 1.863×10^{27} kg)

7. You are designing a roller coaster and want to have the riders feel weightless briefly as they go over the crest of a hill. The hill will have a radius of curvature of 75 m. How fast should the roller coaster be going over this hill in order for the riders to feel weightless? (*Ans*: 27.11 m/s)

8. We often calculate weight by simply using $F_G = mg$, where $g = 9.8$ m/s^2. However, that value of g comes from

$$g = \frac{GM_E}{R_E^2} \tag{5.1}$$

We can calculate weights on other planets or celestial bodies by calculating the value of g on them.

(a) What is the value of g on Mars? (*Ans*: 3.7256 m/s^2)

(b) What is the value of g on the Moon? (*Ans*: 1.6223 m/s^2)

(c) What is the weight of a 72 kg person on Mars, the Moon, and Earth? (*Ans*: 268.24 N, 116.81 N, 705.6 N)

9. **Numerical:** We often say that we can compute the force of gravity for objects near the surface of the Earth with

$$F = mg, \quad \text{where} \quad g = \frac{GM_E}{R_E^2}. \tag{5.2}$$

But, what do we mean by objects near the surface of the Earth? Let's see how the acceleration due to gravity will change as we move away from the Earth's surface. Start by setting up a spreadsheet with a column for the distance above the surface of the Earth, r, starting at 0 and going in steps of 1000 m up to at least 400000 m. Some where off to the side enter the important constants for this problem, the mass of the Earth $M_E = 5.97 \times 10^{24}$ kg, the radius of the Earth, $R_E = 6375000$ m, and Newton's

gravitaitonal constant, $G = 6.67 \times 10^{-11}$ m^2/kg s^2. Then, set up the next column to compute the acceleration due to gravity at a particular height above the Earths surface,

$$g(r) = \frac{GM_\mathrm{E}}{(R_\mathrm{E} + r)^2}. \tag{5.3}$$

Lastly, set up a column to compute the percent difference between the acceleration due to gravity at the surface compared to the distance r above the surface. **Note:** Due to some rounding of things like the mass and radius of the Earth and even the value of G, you will not get 9.8 m/s^2 at the surface. Instead, you shoulf find 9.798 m/s^2.

(a) Create graphs of the acceleration due to gravity as a function of r and the percent diffference with the value at the surface ($r = 0$) as a function of r.

(b) Determine approximately, at what height is the percent difference about 1%, 5%, and 10%.

(c) Planes typically cruise at an altitude around 35,000 feet or 10,700 m. If you dropped an item about 0.5 m from the floor of the air craft at cruising altitude, how much longer would it take to hit the floor than if it were dropped from the same height at the Earth's surface? Is this a significant difference?

(d) Space is generally said to be about 62 miles above the Earth's surface, or 99780 m. By how much does the acceleration due to gravity deviate by the time you enter space?

10. **Numerical:** Two astronauts are floating in space far away from any other gravitating bodies. Both astronauts have a mass of 75 kg. They are separated by 100 m.

(a) Using your knowledge of the force of gravity and how net force causes acceleration, find an expression for the acceleration of one of the astronauts. **Hint:** Set the origin of your coordinate system to the point exactly in the middle of the astronauts, so that one is at some position $-x$ and the other is at some position $+x$. Due to the symmetry of the problem, as the astronauts drift towards each other, the will always be the the same distance as the other one from the origin (e.g. when one is 20 m away, the other is 20 m away on the other side). This means they are separated by $2x$.

(b) Since $G = 6.67 \times 10^{-11}$ N m^2/kg^2, and the masses involved are just 75.0 kg, the force of gravity, and therefore the acceleration, is going to be very small to start. This means that if we were to use a fixed time-step, like we usually do, it will take *a lot* of time-steps for anything to happen. So, for this problem let's get fancy and use an *adaptive time-step*, which just means that we should use one that changes based on what is currently going on in our "simulation." Since the acceleration you found in part (a) should depend on the position, we can use a condition of "constant change in position" to set the value of our time step. Recall from constant acceleration kinematics that the change in position over an interval of time δt is

$$\Delta x = v_0 \delta t + \frac{1}{2} a (\delta t)^2, \quad \text{or} \quad \frac{1}{2} a (\delta t)^2 + v_0 \delta t - \Delta x = 0. \tag{5.4}$$

We can solve this for δt with the quadratic formula,

$$\delta t = \frac{-v_0 \pm \sqrt{v_0^2 - 4(\frac{1}{2}a)(\Delta x)}}{2(\frac{1}{2}a)} \quad \longrightarrow \quad \boxed{\delta t = \frac{-v_0 \pm \sqrt{v_0^2 - 2a\Delta x}}{a}}. \tag{5.5}$$

Now, if we set a fixed Δx, based on the current speed and acceleration, we can calculate a time-step δt that should yield that Δx. When using this expression in solving the problem, note that v_0 refers to the velocity at the previous time-step, and a is the acceleration at that previous time-step (i.e. the values one row up in your spreadsheet).

With all of this information, set up a spreadsheet where the first column is time, the second is position, the third is velocity, the fourth is acceleration, and the fifth is this adaptive time-step. In the first row you with enter a starting time of zero, an initial position for one of the astronauts (how far away from the origin will one of them be at the start?), and the initial velocity will be zero. Enter your expression for acceleration so that it will be calculated from the input values, and then the above expression for the adaptive time-step, letting $\Delta x = 0.01$ m. In the next row below that, let the time be equal to the initial time plus the time-step calculated in that row above. Update the position and velocity as[2]

$$x_1 = x_0 + v_0\delta t + \frac{1}{2}a_0(\delta t)^2, \quad \text{and} \quad v_1 = v_0 + \left(\frac{a_0 + a_1}{2}\right)\delta t. \tag{5.6}$$

Lastly, simply "drag down" the equations you entered to calculate the acceleration and time-step. From here, "drag down" the full row until the astronaut reaches the origin (i.e. $x = 0$ or some really small number like 5.13×10^{-14} m).

(c) Looking at the data in your spreadsheet, how long will it take for the two astronauts to collide with each other? In other words, how long until the astronauts reach the origin?

[2]This way of updating the velocity is what is known as the Velocity Verlet algorithm. It uses the average acceleration over the time-step to update the velocity instead of treating it like the acceleration was constant. This one simple change actually makes a far more robust algorithm compared to our usual method that will give us better results with larger "time-steps."

6 | Work & Energy

1. When moving some boxes around your garage, you can't help but think back to your engineering physics class. You recall the discussions of work and energy, and remember that the net work done on an object moving at a constant speed is zero, so why are you getting so tired sliding all these boxes around. You find that one particular box has a mass of 50 kg, and you estimate that the coefficient of kinetic friction between the cardboard of the box and the concrete floor is $\mu_k = 0.3$.

   (a) Draw and label a free-body diagram for the box as you slide it in a straight line at a constant speed.

   (b) Determine the force you must push the box with to slide it at a constant speed. (*Ans*: 147 N)

   (c) Calculate the work done by each for acting on the box at you move it 10.0 m across the floor. (*Ans*: $W_{\mathrm{N}} = W_{\mathrm{G}} = 0$, $W_{\mathrm{P}} = 1470$ J, $W_{\mathrm{fr}} = -1470$ J)

   (d) Check that the net work done on the box is zero, consistent with the work-energy theorem. (*Ans*: $W_{\mathrm{net}} = 0$)

   (e) Looking at the work done by individual forces, why do you get tired when moving the boxes?

2. You are pulling a wagon ($m_{\mathrm{W}} = 8.0$ kg) with your small cousin ($m_{\mathrm{C}} = 30.0$ kg) sitting in it down a side walk. You are holding the handle at a 20° angle with the horizontal and pulling with a force of 60.0 N.

   (a) Draw and label a free-body diagram for the cousin/wagon system. Assume there is some kind of drag force acting on the wagon (combination of a rolling friction and friction in the axles of the wagon, we don't need to know the exact form of this) that you label as F_{drag}.

   (b) Calculate the work done by all the forces in your free-body diagram, except the drag force, while pulling the wagon 100 m down the sidewalk. (*Ans*: $W_{\mathrm{N}} = W_{\mathrm{G}} = 0$, $W_{\mathrm{P}} = 5638.16$ J)

   (c) How much work must be done by the drag force is the wagon is moving at a constant speed the whole time?

   (d) Using your answer to the previous part, determine the magnitude of the drag force. (*Ans*: 56.3816 N)

3. A block, $m = 2.0$ kg is sitting on an plane inclined at 27° with respect to the horizontal. The coefficient of kinetic friction between the block and the plane is $\mu_{\mathrm{k}} = 0.3$. The block slides a distance of 1.25 m down the incline (Note: this is the distance measured parallel to the inclined surface).

   (a) Draw and label a free-body diagram of the block.

   (b) Calculate the work done by each force on the block as it slides down the plane. Do any of the forces do no work? Why? (*Ans*: $W_{\mathrm{G}} = 11.123$ J, $W_{\mathrm{fr}} = -6.549$ J)

(c) Calculate the net work done on the block as it slides down and use the work energy theorem to determine the speed of the block after sliding the 1.25 m assuming that it starts from rest. (*Ans*: $W_{\text{net}} = 4.574$ J, $v_f = 2.139$ m/s)

(d) Using your free-body diagram as a guide, write out the sum of forces equations and calculate the acceleration of the block. Use this acceleration with the kinematics equations to find the speed of the block after sliding the 1.25 m, again assuming the block starts from rest. (*Ans*: $v_f = 2.139$ m/s)

(e) Do your answers in part c and d agree? Is this expected? Why or why not?

4. You buy a cheap suction-cup dart gun at a dollar store. Looking down the barrel you note that there is a spring that is compressed by pulling back on a handle that sticks out the back, and a small catch that then holds the spring in the compressed state. Pulling the trigger releases the catch, allowing the spring to decompress, propelling the dart forward. You decide to get an estimate of the velocity of the dart as it leaves the gun by firing the dart straight up and timing how long it takes to return to the height of the barrel.

   (a) If the dart takes 4.21 s to travel up and then fall back down to the original height, how fast must it leave the barrel? (*Ans*: 20.629 m/s)

   (b) Using conservation of energy, how high did the dart travel? (*Ans*: 21.712 m)

   (c) Using conservation of energy, and noting that the handle you pull on comes about 5 cm out of the back of the gun when you fully compress the spring, what is the spring constant of the spring inside? Take the mass of the dart to be 20 grams. (*Ans*: 3,404.445 N/m)

   (d) If you want to hit a target that is 10.0 m away and at the same height as the gun when you fire, at what angle should you fire? (*Ans*: 6.657°)

5. An air track glider ($m_{\text{g}} = 150$ g) is at rest on an air track with the air supply turned on (i.e it is on a frictionless surface). A string is tied to the glider, which passes over a pulley. On the other end of this string is a mass hanger ($m_{\text{h}} = 50$ g).

   (a) Draw and label free-body diagrams for the glider and the mass hanger, then write of the sum of forces equations and determine the acceleration of the system. (*Ans*: 2.45 m/s^2)

   (b) Determine how fast the glider (and whole system for that matter) will be moving after the mass hanger has fallen through a height of 50 cm using kinematics equations and the acceleration you just found. (*Ans*: 1.5652 m/s)

   (c) Using conservation of mechanical energy, determine how fast the glider/system will be moving after the mass hanger has fallen a height of 50 cm. (*Ans*: 1.5652 m/s)

   (d) Do your answers agree? Is this expected? Why or why not?

6. You are still working on designing your roller coaster. You know from before that the roller coaster cars need to be traveling $\sim$ 27 m/s at the top of the hill where you want riders to feel weightless briefly. Assuming the cars essentially start from rest to build a bit of suspense, what height difference do you need between the top of the previous hill and the top of the "weightless" hill in order for the roller coaster cars to be traveling at the right speed? Does the exact mass of the roller coaster cars and riders matter? Assume that there is negligible friction and air resistance. (*Ans*: 37.194 m)

7. You are holding a softball, $m = 0.2$ kg, 1.0 m above a flat plate attached to a spring with spring constant $k = 431$ N/m.

(a) What is the speed of the ball when it hits the flat plate? (*Ans*: 4.427 m/s)

(b) What distance will the spring compress? (*Ans*: 0.100 m)

8. The most fuel efficient way to get a satellite into a geosynchronous orbit (i.e. one with an orbital period equal to one *sidereal* day, $P = 23^{\text{h}}56^{\text{m}}4.0905^{\text{s}}$) is through a Hohmann transfer orbit. This involves first launching the satellite into a low-Earth orbit (about 300 km above the surface), then firing a booster so that the satellite gets up to a higher speed putting it on an elliptical orbit. When the satellite then reaches the desired distance for the geosynchronous orbit, the booster fires again, increasing the satellites speed so that it assumes the circular geosynchronous orbit. Let's examine the energy needed for this to be done.

   (a) Using the period of the geosynchronous orbit, compute the radius of the orbit and the speed the satellite will have when in that orbit. (*Ans*: 4.215×10^7 m)

   (b) Calculate the speed that the satellite will have in its low-Earth orbit. (*Ans*: 7721.89 m/s)

   (c) Calculate the gravitational potential energy of the 1000 kg satellite in each orbit. (*Ans*: $U_{\text{GS}} = -9.447 \times 10^9$ J, $U_{\text{LE}} = -5.963 \times 10^{10}$ J)

   (d) Calculate the kinetic energy of the satellite in each orbit. (*Ans*: $K_{\text{GS}} = 4.724 \times 10^9$ J, $K_{\text{LE}} = 2.981 \times 10^{10}$ J)

   (e) Determine the amount of work that the booster rocket will need to do on the satellite to get it into its geosynchronous orbit. (*Ans*: 2.509×10^{10} J)

9. You have a bouncy ball and are trying to determine how much energy is lost on each bounce. The ball has a mass of 50 grams, and you drop it from a height of 1.00 m. Through the use of a high-speed camera, you determine that after the first bounce, the ball comes back to a height of 0.8 m.

   (a) What fraction of the energy is lost after this first bounce? What fraction of energy is still in the system? (*Ans*: 20% lost, 80% remains)

   (b) Assuming the same fraction of energy is lost on each bounce, what would you expect the maximum height of the ball to be after the second bounce? The third? The fourth? (*Ans*: 0.64 m, 0.512 m, 0.4096 m)

   (c) Derive an expression for the energy of the bouncy ball after some number n bounces in terms of the fraction of the energy that remains after a bounce, e, and the initial energy of the bouncy ball, E_0. **Hint:** Write E_1 in terms of E_0, then write E_2 in terms of E_1 then substitute your expression for E_1 in terms of E_0. Keep doing this for subsequent energies until you notice a pattern that allows you to write E_n in terms of e, n, and E_0.

   (d) Using your expression, how many times will the ball bounce before effectively coming to rest? Meaning, how many bounces until the energy of the ball is $10^{-6}E_0$? (*Ans*: About 62 bounces)

10. **Numerical:** Examining a problem based on energy can often (but not always) make the problem much simpler to solve then analyzing forces and using kinematics. Either approach we take should give us the same answers, otherwise there is something wrong with our understanding of the physical world. Consider a block, $m_{\text{A}} = 5.0\ kg$ sitting on a plane inclined at 30° to the horizontal. The coefficient of kinetic friction between the block and the plane is $\mu_{\text{k}} = 0.25$. A string is tied to block A and then runs over a pulley where it is attached to block B, $m_{\text{B}} = 5.0\ kg$, which is hanging freely.

(a) Draw and label free-body diagrams for both blocks, write out the sum of forces and use those equations to determine the acceleration of the block. Is this acceleration constant?

(b) Using the general conservation of energy, derive an expression for the speed of the system as a function of the distance block B has fallen. Note that the distance block A must rise can be related to the distance block B has fallen through trigonometry, and the distance along the incline that block A will have moved is always equal to the distance block B has fallen.

(c) In a spreadsheet, use the acceleration you found in part a to calculate the distance block B will have fallen, and the velocity of the system as functions of time between zero and 1.2 seconds in steps of 0.1 s. Since the acceleration is constant here, we can just use the kinematics equations normally. Assume the system starts at rest.

(d) In another location in the spreadsheet, calculate the velocity of the system as a function of the distance block B has fallen through using your expression from the energy analysis. Make a column for the distance that starts at zero, then goes in steps of 0.025 m until it gets to 0.1 m, then go in steps of 0.1 m. Again, we don't have to do anything funny with time steps here, just have the spreadsheet compute the values using your derived expression.

(e) Note that since

$$v = v_0 + at, \quad \text{and} \quad v^2 = v_0^2 + 2a\Delta y_B \tag{6.1}$$

where $v_0 = 0$ and Δy_B is the distance block B has fallen. We can find that

$$t = \frac{2\Delta y_B}{v}. \tag{6.2}$$

Using this expression, compute the time at which block B will have fallen a given distance from the energy analysis data.

(f) Create a graph that plots the distance block B has fallen as a function of time from the two different data sets. Have Excel draw smooth lines through the data points and make sure both data series are on the same graph. Do the lines agree with each other? Is this expected? Why or why not?

7 | Momentum

1. A block, $m_A = 1.0$ kg is sliding on a frictionless surface at $v_A = 5.0$ m/s towards a second block $m_B = 2.0$ kg that is initially at rest.

   (a) Assuming that the collision is elastic, what are the velocities of the blocks after the collision? (*Ans*: $-1.\overline{66}$ m/s, $3.\overline{33}$ m/s)

   (b) If the blocks are in contact during the collision for 1×10^{-3} s, what is the magnitude of the average force of the blocks on each other? (*Ans*: 6,667 N)

   (c) Assuming that the collision is completely inelastic, what are the velocities of the blocks after the collision? (*Ans:* $1.\overline{66}$ m/s)

2. A particle of mass m_1 is heading towards a particle of mass m_2 that is initially at rest. The two particles collide head-on (i.e. this is a one-dimensional problem) and after the collision, the first particle is observed to be moving in the opposite direction with a speed of 0.3 times it's original speed. Assume the collision is elastic.

   (a) Find an expression for the mass of the second particle in terms of the mass of the first particle. (*Ans*: $m_2 = 1.857m_1$)

   (b) Find an expression for the velocity of the second particle after the collision in terms of the mass of the first particle and its initial velocity. (*Ans*: $v_2' = 0.7v_1$)

3. You hang two steel balls, $m_A = 0.050$ kg and $m_B = 0.075$ kg from 45 cm long pieces of string so that they just touch when both hang vertically at rest. You pull ball A back at a 60° angle with respect to the vertical and let it go, while ball B remains hanging vertically at rest.

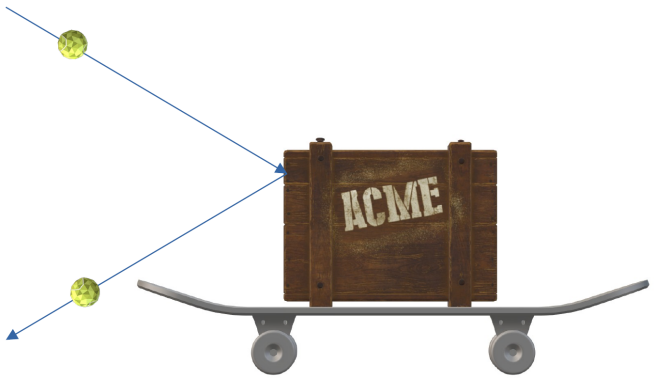
   (a) Ball A is let go. At what velocity will ball A impact ball B? (*Ans*: 2.1 m/s)

   (b) Assuming that the collision is elastic, what will the velocities of the balls be after the collision? (*Ans*: -0.42 m/s, 1.68 m/s)

   (c) How high will ball A go after the collision? What angle will ball A's string make with the vertical? (*Ans*: 0.009 m, 11.478°)

   (d) How high will ball B go after the collision? What angle will ball B's string make with the vertical? (*Ans*: 0.144 m, 47.156°)

4. As a simple at home momentum experiment, you take a crate ($M_C = 16$ kg) and place it on a skateboard ($m_S = 2.0$ kg). The crate/skateboard system is initially at rest. You then throw a tennis ball ($m_b = 0.057$ kg) at the crate at a speed of 30.0 m/s and an angle of 20° below the horizontal. The tennis ball bounces off the crate at a 30° angle below the horizontal (see the diagram below).



(a) Write out the conservation of momentum equations for this collision in both the x and y-directions.

(b) Using your equation for the y direction, what is the speed of the tennis ball after the collision? (*Ans*: 20.521 m/s)

(c) Using your equation for the x direction, what is the speed of the crate/skateboard system after the collision? (*Ans*: 0.146 m/s)

(d) Is this an elastic or inelastic collision? How do you know?

5. In major league baseball, a fastball can be thrown with speeds of 90 mph (40.23 m/s) on average. After being hit, the ball can be traveling around 130 mph (58.12 m/s). A pitch that was traveling in the negative direction at 90 mph, 5° below the horizontal, is hit and is then traveling at 130 mph in the positive direction, 20° above the horizontal. The mass of a baseball is 145 grams.

   (a) Find the initial momentum of the baseball before it is hit. Report this in unit vector notation. (*Ans*: $\vec{\mathbf{p}}_i = (-5.8116\,\hat{\mathbf{i}} - 0.5084\,\hat{\mathbf{j}})$ kg m/s)

   (b) Find the final momentum of the baseball after it is hit. Report this in unit vector notation. (*Ans*: $\vec{\mathbf{p}}_f = (7.919\,\hat{\mathbf{i}} + 2.882\,\hat{\mathbf{j}})$ kg m/s)

   (c) What is the impulse felt by the baseball while being hit? Report this in unit vector notation. (*Ans*: $\vec{\mathbf{J}} = (13.731\,\hat{\mathbf{i}} + 3.390\,\hat{\mathbf{j}})$ kg m/s)

   (d) Assuming that the bat is in contact with the baseball for 3 milliseconds (3.0×10^{-3} s), what is the average force applied to the baseball by the bat? Report this in unit vector notation. (*Ans*: $\vec{\mathbf{F}}_{\text{avg}} = (4577\,\hat{\mathbf{i}} + 1130\,\hat{\mathbf{j}})$ N)

6. A satellite is heading towards the Kuiper belt to study the objects located there. At a certain point in its journey, the heading of the satellite needs to be adjusted by 21° to avoid being pulled in by Neptune's gravity. The satellite uses gas maneuvering thrusters that shoot out stored gas at a high speed, 2000 m/s, in order to change the trajectory. The mass of the satellite and gas in the tanks is 3496 kg, and it is initially traveling at 120 m/s. Assuming that the thruster is fired briefly so that the gas is expelled perpendicular to the original direction of the motion, how much mass must be ejected to alter the heading of the satellite? (*Ans*: 80.52 kg)

7. You get a job as a forensic investigator right out of college. Since you are the new person, you are assigned the task of measuring the velocities of bullets fired from various guns. To do this, you need to build a testing apparatus and decide to use your engineering physics knowledge to help you. You affix a spring (spring constant $k = 150$ N/m) to a wall at a height so that it can line up with a large

block of wood (M_B) placed on a table and rested up against the uncompressed horiztontal spring. When a bullet (m_b) is fired it will embed into the wood block in a completely inelastic collision. The coefficient of kinetic friction between the wood block and the table is $\mu_k = 0.25$.

(a) Using conservation of momentum, derive an expression for the speed of the block-bullet system after the bullet completely embeds into the block in terms of the mass of the bullet, the mass of the block, and the initial speed of the bullet.

(b) Using conservation of energy, derive an expression for the initial speed of the bullet in terms of the masses, the spring constant, and the compression of the spring.

(c) A particular gun fires bullets with mass $m_b = 0.001$ kg. You place a block of wood with a mass of $M_B = 0.999$ kg at rest up against the spring, but not compressing it. You observe that the spring compresses a maximum of 0.050 m. What is the speed of the bullet when it leaves the gun? (*Ans*: 787.4 m/s)

8. After a snow storm, you are once again stuck watching your younger cousin as their school is closed for the day. You take them sledding with your cousin sitting on your lap as you both sled down the hill. After going down the hill, there is a patch of ice that lets the sled slide without friction for a short time. On your 20th or so time down the hill, you decide that you've had enough of your younger cousin and your throw them off the sled perpendicular to direction of travel as viewed by an observer at rest with respect to the ground. If you have a mass of 75 kg, your cousin has a mass of 25 kg, and the plastic sled has a mass of 2 kg, you were originally traveling at 10 m/s, and your throw your cousin off at 1 m/s, what is your velocity after throwing your cousin? (*Ans*: $\vec{\mathbf{v}}_f = (13.25\,\hat{\mathbf{i}} + 0.325\,\hat{\mathbf{j}})$ m/s)

9. A particle of mass $m = 0.25$ kg is subjected to a time varying force,

$$\vec{\mathbf{F}}(t) = (3.0t\,\hat{\mathbf{i}} - 0.5t^2\,\hat{\mathbf{j}})\,N, \tag{7.1}$$

for a period of 3.0 s.

(a) What is the impulse felt by the particle? (*Ans*: $\vec{\mathbf{J}} = (13.5\,\hat{\mathbf{i}} - 4.5\,\hat{\mathbf{j}})$ kg m/s)

(b) If the particle starts at rest, what is it's velocity after the force has acted on it? (*Ans*: $\vec{\mathbf{v}}_f = (54\,\hat{\mathbf{i}} - 18\,\hat{\mathbf{j}})$ m/s)

(c) Using your knowledge of Newton's Laws and the relationship between velocity and acceleration, verify your answer in the previous part.

10. **Numerical:** Collisions in two dimensions can be quite difficult to analyze analytically (i.e. it's hard to derive equations that describe them). However, with the conservation of momentum, it can be fairly easy to put things together on a computer. Consider two billiards balls colliding with each other. Assume that ball A is initially moving at 8.0 m/s in the x-direction, and ball B is initially at rest. After the collision, assume that ball A moves off at an angle θ_A and that ball B moves off at an angle $\theta_B = -\theta_A$. The angle between the final velocities is then $\theta_{AB} = 2\theta_A$.

(a) Derive expressions for the magnitude of the final velocities of both balls, v'_A and v'_B in terms of magnitude of the initial velocity of ball A, the masses of the balls, m_A and m_B, and the angle between the final velocities θ_{AB}.

(b) Set up a spreadsheet with the first column containing different θ_{AB} angles, starting at 10° and going to 120°. Then, using your expressions from part (a), set up the second and third columns to be able to compute the velocities of the balls after the collision. You will likely want to put down the initial velocity of ball A and the masses of the balls in some space off to the side so you can reference those cells in your formulas as needed. For now, assume that both balls have the same mass, i.e. $m_A = m_B = 0.17$ kg. In the fourth column, find the kinetic energy of the balls after the collision, and under the masses and velocities of the ball, compute the initial kinetic energy. In the fifth column, find the change in kinetic energy ($\Delta K = K_f - K_i$).

(c) Scroll through your data after you fill in the cells for all of the angles and find the angle at which the change in kinetic energy is zero. What angle is this? Based on your knowledge of energy, which angles are impossible?

(d) It turns out that the mass of billiards balls is slightly different than the mass of the cue ball. Assuming that ball A is the cue ball and has a mass $m_A = 0.16$ kg, and ball B has a mass of $m_B = 0.17$ kg. What angle is the change in kinetic energy zero now? What are the impossible angles?

8 | Rotational Motion & Angular Momentum

1. Old-fashioned hard-disk drives for computers use spinning platters that can be magnetized to store data. Some of these drives spin their platters at 7200 rpm, and take 3.0 s to reach that speed from rest.

   (a) Convert rotational frequency, 7200 rpm, to the angular velocity in radians per second. (*Ans*: 753.98 rad/s)

   (b) Assuming that the platters are spun-up uniformly from rest, what is the angular acceleration? (*Ans*: 251.33 rad/s^2)

   (c) What will the angular velocity be at 1.5 s? (*Ans*: 376.99 rad/s)

   (d) What angle will the platters have spun through by the time they reach their top angular speed? (*Ans*: 1130.97 rad)

2. A uniform rod of length ℓ and mass m_{R} is sat on a pivot and held so that it remains horizontal. The pivot is located a distance of $\ell/3$ from the left end of the rod. Two disks of mass m_{D}, and negligible thickness, have holes in the middle and are slid onto the rod, one at each end. Find an expression for the net torque on the rod the instant that the rod is released in terms of the masses of the disks and rod, and the length of the rod. Why is this only the torque for the instant that the rod is released? (*Ans*: $\sum \tau = -g\ell[m_{\mathrm{D}} + m_{\mathrm{R}}/2]/3$)

3. A 12-inch (diameter) vinyl record has a mass of 150 grams. The moment of inertia of a disk is

$$I = \frac{1}{2}MR^2. \tag{8.1}$$

   These records are typically spun at 33 and 1/3 rpm while playing, and take about 2.0 seconds to reach that rotational speed.

   (a) Assuming that the disk spins up uniformly, what is the angular acceleration of the disk? (*Ans*: 1.745 rad/s^2)

   (b) What torque must the motor in the record player apply to the disk during this period of time? (*Ans*: 3.04×10^{-3} N $\cdot$ m)

4. A 1.0 kg ball of clay (uniform density sphere) with a radius of 3.8 cm is sitting on an electric potters wheel. When it was thrown onto the wheel, it landed a little off center and is sitting with its center located 2.0 cm from the center of the wheel. The wheel can be modeled as a disk of mass 6.0 kg and radius of 35 cm.

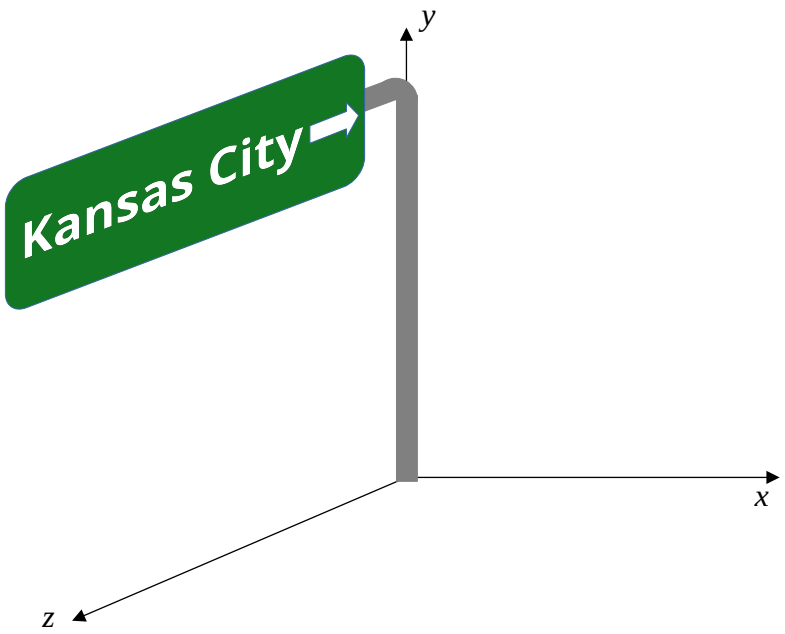
(a) Find the moment of inertia for the wheel/clay ball system. (*Ans*: 0.3685 kg m^2)

(b) Pottery wheels can spin at rates up to 300 rpm to make shaping of the objects easier. If the wheel starts from rest and reaches 300 rpm in 30 seconds, what is the angular acceleration of the system? (*Ans*: $\pi/3$ rad/s^2)

(c) What is the torque that the motor must supply? (*Ans*:0.3859 N·m)

(d) The motor is turned off. If there is a frictional torque in the bearings of $\tau_{\text{fr}} = -0.0643\ \text{N} \cdot \text{m}$, how long will it take for the wheel to come to rest naturally? How many times will it have rotated when it does come to rest? (*Ans*: $t = 180.0$ s, 450.1 revolutions)

5. After graduating, you get a job with a construction company that has been charged with putting up a new road sign. In order to figure out what bolts to use to secure the post to the concrete anchor, you need to calculate the maximum expected torque the sign might produce at the anchor point (i.e. the origin in the diagram below) due to the weight of the sign and support arm, as well as the maximum force expected due to winds. The center of mass of the sign and support arm is located at $\vec{\mathbf{r}}_{\text{CM}} = (6.10\,\hat{\mathbf{j}} + 9.00\,\hat{\mathbf{k}})$ m, and the sign plus the horizontal support have a combined mass of 500 kg. Based on historical data, you expect the maximum force from the wind to be $\vec{\mathbf{F}}_{\text{wind}} = \pm 2600\ \text{N}\,\hat{\mathbf{i}}$. What is the maximum torque about the anchor point? (*Ans*: $\vec{\tau} = [4.41 \times 10^4\,\hat{\mathbf{i}} \pm 2.34 \times 10^4\,\hat{\mathbf{j}} \mp 1.586 \times 10^4\,\hat{\mathbf{k}}]$ N·m)



6. Angular momentum is conserved unless there is a net external torque of the system. The angular momentum of a particle about the origin of a coordinate system can be computed as

$$\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}} = m\vec{\mathbf{r}} \times \vec{\mathbf{v}}. \tag{8.2}$$

A free-particle, by definition, has no forces acting on it, and therefore cannot have any torques acting on it. That also means, according to Newton's first law, that the particle will move in a straight line at a constant speed. So, is the angular momentum of a free-particle conserved about an arbitrary origin?

(a) Start by picking a coordinate system such that the particle, of mass m, is moving only in the x-direction at speed v_x. This means that it is at some fixed distance away from the x-axis that we will simply call y, and has a time dependent x position given by $x(t) = x_0 + v_x t$, since v_x is constant because there is no net force and therefore no acceleration. From this, derive an expression for the angular momentum of the particle. Does your expression depend on time?

(b) Next, find the angular momentum for this particle with a completely arbitrary choice of coordinate system. That is, the constant velocity of the particle is

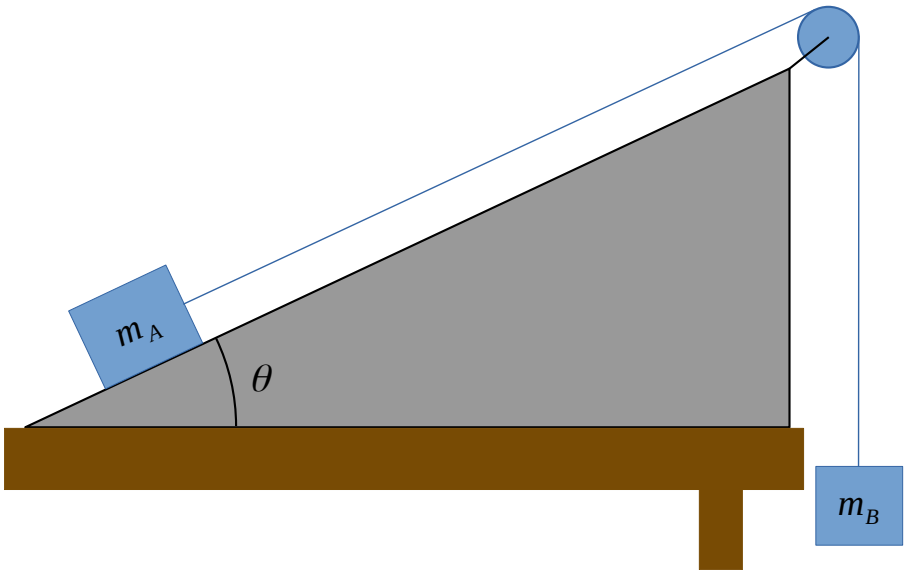
$$\vec{\mathbf{v}} = v_x\,\hat{\mathbf{i}} + v_y\,\hat{\mathbf{j}} + v_z\,\hat{\mathbf{k}}, \tag{8.3}$$

and the time dependent position of the particle is

$$\vec{\mathbf{r}}(t) = (x_0 + v_x t)\,\hat{\mathbf{i}} + (y_0 + v_y t)\,\hat{\mathbf{j}} + (z_0 + v_z t)\,\hat{\mathbf{k}}. \tag{8.4}$$

Using the vectors given above, derive an expression for the angular momentum of this particle. Does your expression depend on time?

7. When dealing with pulleys in the past, we have always assumed that it was an ideal pulley in the problem, meaning that is was massless and frictionless. Of course, in the real world, pulleys have mass and will also have some friction in their bearings, but with good bearings, this friction can still be negligible. Consider the situation shown in the diagram below where $m_A = 2.0$ kg, $m_B = 1.7$ kg, $\mu_k = 0.3$, and $\theta = 25.0°$.

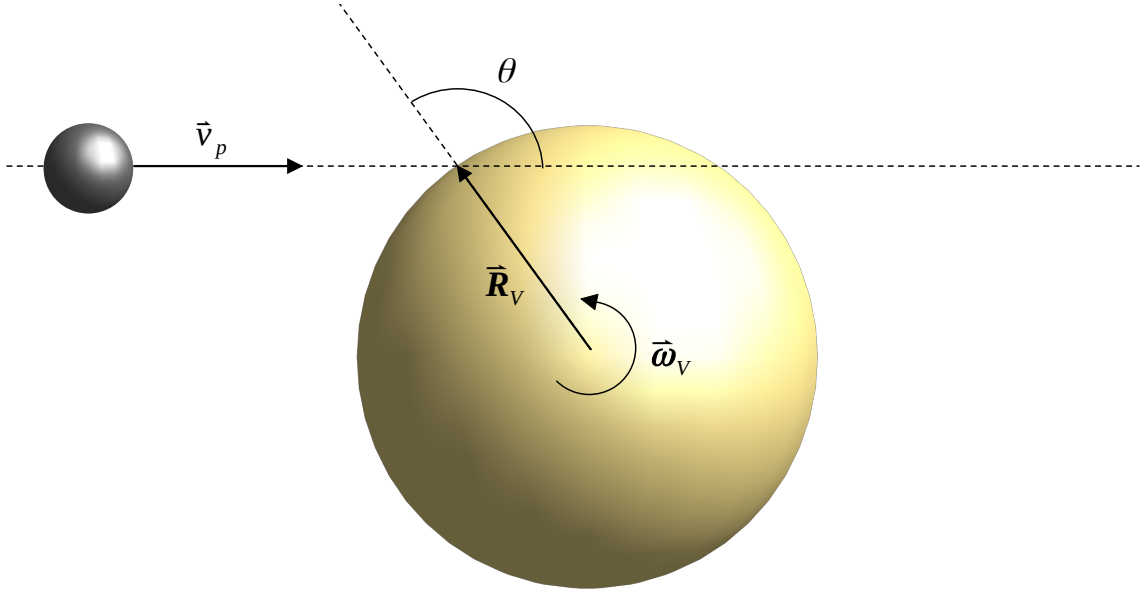


(a) Draw and label free-body diagrams for both masses in the case of an ideal pulley. Using those diagrams as a guide, write out the sum of forces equations and derive an expression for the acceleration of the system.

(b) Draw and label free-body diagrams for both masses and an extended free-body diagram for the pulley for the case that the pulley is modeled as a uniform disk of mass $m_p = 0.200$ kg and radius $R_p = 5.0$ cm. Using the diagrams as a guide, write down the sum of forces equations and the sum of torque equation for the pulley and derive an expression for the acceleration of the system. **Hint:** In this case, the pulley needs to rotate so there will be a net torque on the pulley which means the tension won't be the same everywhere in string connecting the masses.

(c) Using the given numbers, compute the values of the acceleration in each case. If you were to ignore the mass of the pulley, what would the percent error be in the acceleration you calculate? (*Ans*: 2.70%)

8. Venus is a bit of an odd ball planet in that it rotates (spins on its axis) in the opposite direction compared to most of the other planets in our solar system. One of the theories as to how this happened is that a large object collided with Venus in the past causing its rotation to reverse or for Venus to flip upside down. Let's examine the case that the impact caused the rotation to reverse. Currently, it takes Venus 243 days to rotate once as measured relative to distant stars, but it does so in the opposite direction (call this the negative direction). Assume that initially Venus rotated in the positive direction at about the same rate as Earth, once every 24 hours. If Venus was struck by a large object of mass $m_p = 9 \times 10^{20}$ kg, which impacted at Venus's equator as shown below ($\theta = 120°$) and embeded itself into Venus's surface at the point of impact, at what speed would the impact have had to happen to reverse Venus's rotation? Note that the mass of Venus is $M_V = 4.87 \times 10^{24}$ kg, and its radius is $R_V = 6.05 \times 10^6$ m. (*Ans*: 1.104×10^6 m/s)



9. You have three, round objects, a uniform density sphere, a uniform density disk, and a uniform density ring. You place these three objects onto an inclined plane such that they all start at a height h above the flat horizontal surface that the incline rests on. The objects are released and they roll without slipping down the incline.

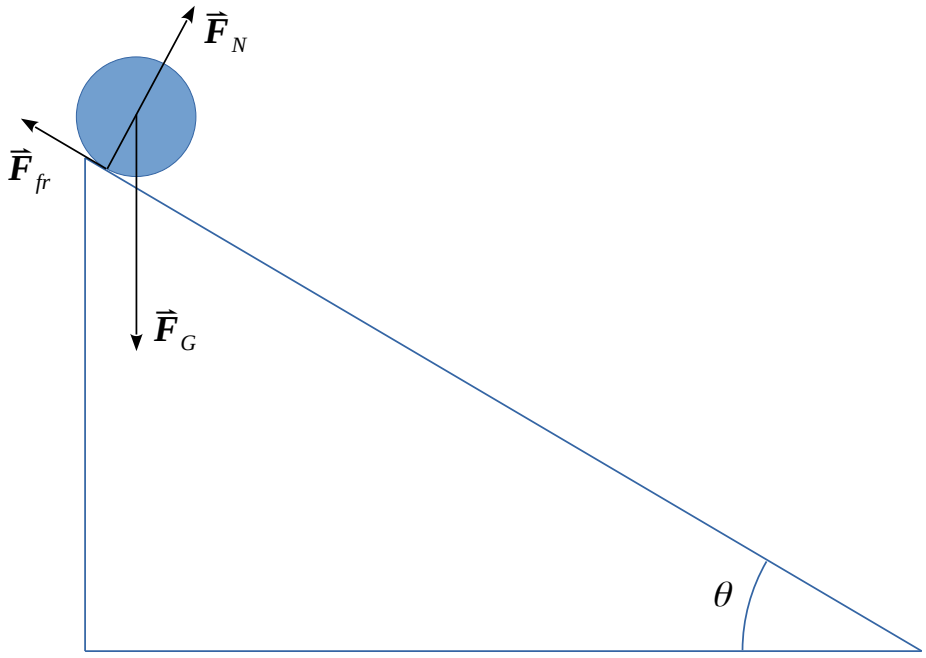
   (a) Using conservation of energy, derive expressions for the speed of the objects when they get to the bottom of the ramp (i.e. when their height is zero).

   (b) Do your expressions depend on the mass or radius of the objects?

   (c) Which object is going the fastest at the bottom? Which object gets their first?

10. **Numerical:** In the previous problem you computed the speed of various round objects that would roll without slipping down an inclined plane. As part of that you were asked which objects were moving the fastest at the bottom and which got there first. While the object that is moving the fastest is relatively easy to see, which one gets there first can be a bit harder. In this numerical problem, let's

create some graphs that show the positions of the objects as a function of time and their velocities as a function of time as well. So, we have a round object that rolls without slipping down the inclined plane. The extended free-body diagram of this object is as shown below.



(a) Using the diagram above as a guide, write down the sum of forces equations in the x and y-directions, and write down the sum of torques about the center of the round object. For now, just use m for the mass, I for the moment of inertia, and R for the radius of the object. Then, use the equations to find an expression for the acceleration in terms of the mass of the object, it's moment of inertia, the angle of the incline, θ, and the radius of the object. Is this acceleration a constant?

(b) Using the formulas for the moments of inertia for a uniform density sphere, disk, and ring, write out the accelerations for each. Do you need to know the mass or the radius of the objects?

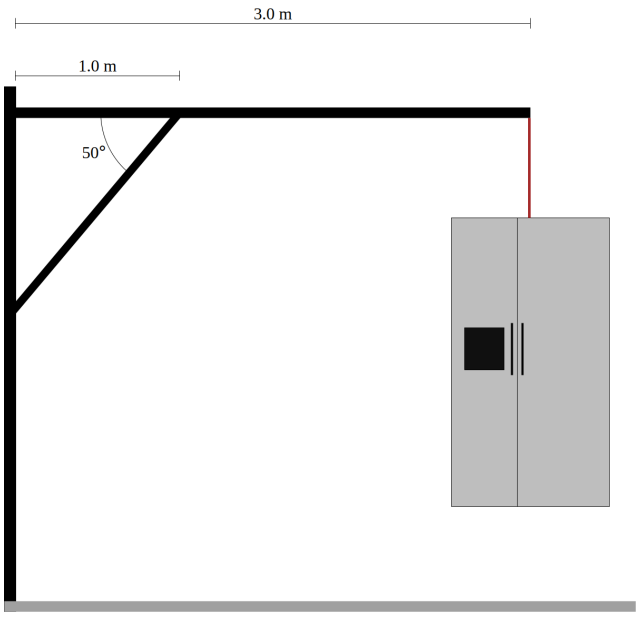
(c) Using your constant acceleration kinematics equations and your knowledge of trigonometry, derive an expression that will give you the time it will have taken for the rolling object to have had a particular change in height.

(d) Assume that the objects start at a height of 1.0 m above the flat, horizontal plane that the incline is resting on. All the objects start from rest, and the angle of the incline is $\theta = 30.0°$. Set up a spreadsheet with the first column being a particular height above the horizontal surface, starting at 1.0 m and going to zero in steps of 0.01 m. Then, in the next three columns, use your expression for the time it will take for a particular object to have a change in height corresponding to the the height in the first column, for example, the objects start at 1.0 m, and then the next height is 0.99 m, so the change is height is 0.01 m. This means you can compute the change in height as the difference between the initial height and the current height. Lastly, use your expressions from the previous problem to compute the speeds of the objects at the various heights.

(e) Create a graph of the height as a function of time for each of the objects. These graphs should then clearly show which object reaches the bottom of the ramp the fastest. Also, create graphs of the velocities as a function of time.

9 | Static Equilibrium – Stress – and Fracture

1. You are working on a home construction site over summer break to make some cash. You and your coworkers get bored while eating lunch and decide to set up a makeshift balance beam to see who can walk the full length. A 10 foot long (3.00 m), 9.0 kg, 2"×6" is supported on its ends by saw horses. You ($m = 76.0\ kg$) are walking across the board and have made it 1/3 of the way along the beam.
   (a) Draw and extended free-body diagram of the board.
   (b) Write out the sum of forces and a sum of torque equation cleverly picking your axis of rotation so that one of the unknown forces doesn't factor in.
   (c) What are the upward forces provided by the saw horses? (*Ans*: 292.37 N, 540.63 N)
   (d) Pick an axis of rotation at the center of the beam and sum the torques about that axis. Pick an axis of rotation at 2/3 the length of the beam and sum the torques about that axis. In a static problem, does our choice of axis of rotation effect the sum of the torques? **Hint:** Work in terms of variables, not numbers, as rounding may effect your conclusions.
2. While moving you construct a rudimentary lift to help you lift some heavy things onto the truck. It consists of a horizontal, uniform beam ($m_{\text{beam}} = 20.0$ kg) supported by a strut that connects at a 50.0° angle at 1/3 the length of the beam as shown below. There is a cable that can be raised and lowered to lift the loads. You are using it to lift your 300 lbs (136.0 kg) fridge.



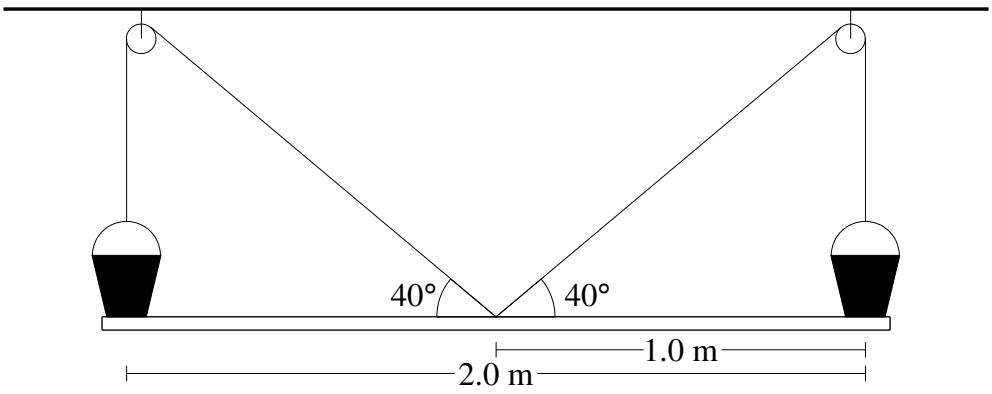
(a) Draw and extended free-body diagram of the horizontal beam.

(b) With the fridge lifted off the ground, what is the magnitude of the force from the strut on the beam? (*Ans*: 5603 N)

(c) What is the vertical force from the supporting pole (vertical part all the way to the left) on the horizontal beam? **Hint:** Sum the forces in the y-direction. (*Ans*: -2763 N)

(d) What is the horizontal force from the supporting pole (vertical part all the way to the left) on the horizontal beam? (*Ans*: -3602 N)

3. A uniform plank of mass $M_\text{p} = 12$ kg has two ropes connected to its center. These ropes go up to two pulleys at an angle of 40° to the horizontal. They then go down vertically to where they are attached to two buckets of the same mass $m_\text{b} = 10$ kg. The buckets also rest against the plank (i.e. there is a normal force, F_N between each bucket and the plank). See the diagram below. **Note:** The whole system is completely at rest.



(a) Draw a free-body diagram for one of the buckets and write out the sum of forces on the bucket. Would anything be different for the other bucket?

(b) Draw an extended free-body diagram for the plank. Sum the forces and the torques (about the center) for the plank. What should those equations be equal to? Which one is the only one that matters?

(c) What is the tension in the string? (*Ans*: 95.45 N)

(d) What is the normal force between the plank and a bucket? (*Ans*: 2.55 N)

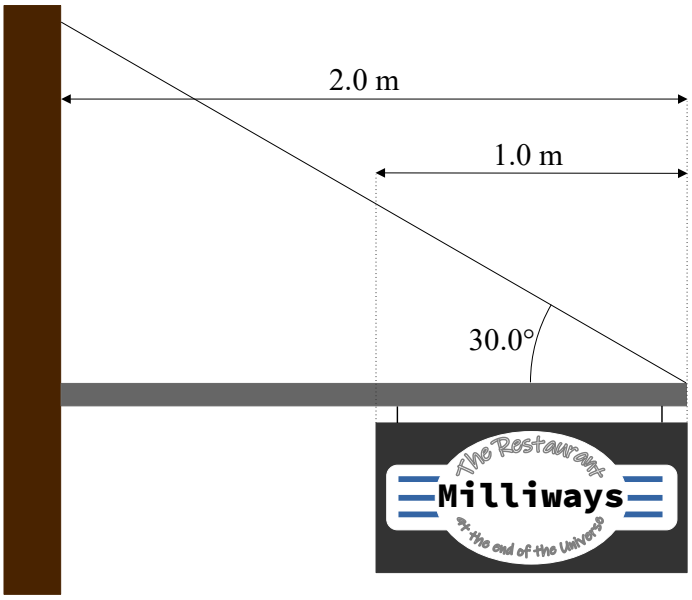
4. You are hired by a new business that just opened a downtown location and would like to hang a somewhat old-fashioned sign from the wall outside. The sign has a mass of $m_\text{sign} = 25.0$ kg, and is 1.0 m square. The pole that you will hang it from is 2.0 m long and has a mass of 10.0 kg. You attach a steel cable at a 30.0° angle to help support the end of the beam. Both the cable and the beam are attached to the wall.

(a) Draw and extended free body diagram of for the beam, splitting the force at the wall into a vertical and horizontal component. **Note:** The force from the sign can be treated as a single force acting on the beam at the point that lines up with the sign's mid-point.

(b) What is the tension in the cable? (*Ans*: 465.5 N)

(c) What is the horizontal force from the wall on the beam? (*Ans*: 403.13 N)

(d) If the tensile strength of the steel cable is 5.00×10^8 N/m^2, what is the minimum cross-sectional area of the cable needed to support the beam and sign? (*Ans*: 9.31×10^{-7} m^2)



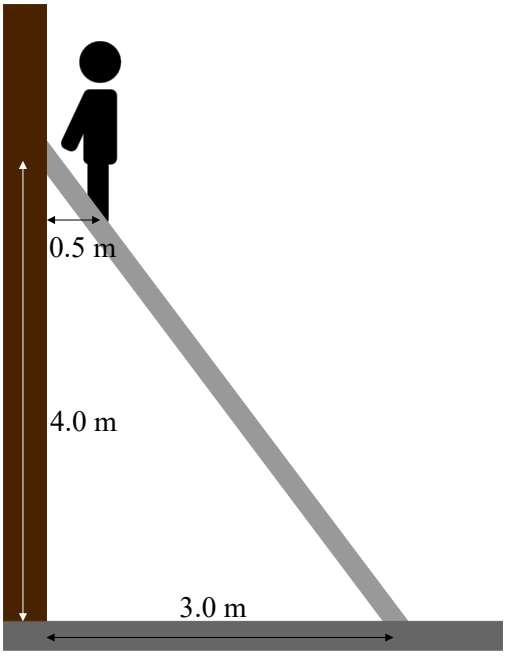
5. As you are going to hang the sign mentioned above, you need to climb up a ladder to mount some brackets. You lean the 5.0 m long ladder ($m_L = 20.0$ kg) against the wall so the top of the ladder is 4.0 m up the wall, and the base is 3.0 m from the wall (see the diagram below). You climb the ladder to a point where you are 0.5 m from the wall. Assume that there is only friction between the ground and the base of the ladder, but no friction between the wall and the top of the ladder, and that you have a mass of 76.0 kg.

   (a) Draw an extended free body diagram of the ladder.

   (b) What is the normal force from the wall on the ladder? (*Ans*: 539 N)

   (c) What is the normal force from the ground on the ladder? (*Ans*: 940.8 N)

   (d) What is the smallest coefficient of friction between the ground and the ladder that would prevent the ladder from sliding? (*Ans*: 0.573)



6. Venus has an extremely dense atmosphere, with a surface pressure of 9.3×10^6 Pa (about 93 times greater than the surface of the Earth. If you sent an aluminum cube ($B = 70 \times 10^9$ N/m^2) from Earth to Venus, by what fraction would its volume change under the immense pressure on Venus' surface? (*Ans*: -1.314×10^{-4})

7. You have been hired to hang a chandelier in the lobby of an upscale hotel. The chandelier has a mass of 225.0 kg, and is to be hung with a steel cable, which has a tensile strength of 5.00×10^8 N/m^2 and and elastic modulus of $E = 2.00 \times 10^{11}$ N/m^2, that needs to have a thickness such that there is a safety factor of 9 (i.e. the cable can withstand 9 times the normal load of the chandelier).

   (a) What is the needed diameter of the cable? (*Ans*: 0.711 cm)

   (b) If the chandelier is to hang down 2.0 m from the ceiling, how long should the cable be before the chandelier is hung from it? (*Ans*: 1.999 m)

8. Going back to the problem where you where hanging the sign. How long should the steel cable, with an elastic modulus of $E = 2.00 \times 10^{11}$ N/m^2, be so that when under the load the beam will be horizontal? Assume that the cable has the cross-sectional area you found in that previous problem. (*Ans*: 2.308 m)

9. Consider the makeshift balance beam from a previous problem. The shear strength of pine is 5×10^6 N/m^2, will the board break as you try to cross it? **Hint:** What is the maximum shear force for this board?

10. **Numerical:** After finishing your degree, you are working for a company that develops new metal alloys to be used in construction. You are given a sample of one of these new alloys in the form of a wire with a diameter of 1.00 cm and a length of 5.0 cm, and asked to find its elastic modulus. You put it on a piece of test equipment that stretches the wire and measures the amount of force needed to stretch the wire by different ammounts, and you obtain the following data:

Load (kN)	Elongation (cm)
0	0
1.5	0.0005
4.6	0.0015
8	0.0025
11	0.0035
11.7	0.005
11.8	0.008
12	0.02
16.6	0.04
20	0.1
21.5	0.28
19.5	0.4
18.5	0.46

(a) Graph the stress on the wire versus the strain on the wire.

(b) Identify the elastic region of the graph.

(c) Using only the elastic region, determine the elastic modulus of this alloy. (*Ans*: 2.018×10^{11} N/m^2)

10 | Fluids & Buoyancy

1. We tend to make simplifying assumptions in physics to make calculations easier. One such assumption would be to treat the Earth ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6378.1$ km) as a uniform density sphere when computing its moment of inertia. If we wanted to make a slightly more realistic assumption, we could instead divide the Earth up into three regions with different densities: an inner core from 0 to 1220 km with a density of 13,000 kg/m^3, an outer core from 1220 to 3480 km with a density of 11,100 kg/m^3, and then a mantle from 3480 km to the surface (the crust is very thin compared to these layers, so we ignore it) with a density of 4,400 kg/m^3.

   (a) Using this model, find the average density of the Earth. (*Ans*: 5,503 kg/m^3)

   (b) Using the mass and radius of the Earth, find the actual average density and compare this to what you found above using a percent difference. (*Ans*: 5,493 kg/m^3, 0.18% difference)

   (c) Using the model, calculate the moment of inertia of the Earth. Note that a spherical shell with radius r_2 and cavity with radius r_1 has a moment of inertia of

   $$I = \frac{2}{5} m \left(\frac{r_2^5 - r_1^5}{r_2^3 - r_1^3} \right). \tag{10.1}$$

   (*Ans*: 8.355×10^{37} kg $\cdot$ m)

   (d) Modeling the Earth as a sphere of uniform density, calculate its moment of inertia and compare to that of the three layer model. (*Ans*: 9.714×10^{37} kg $\cdot$ m, 16.3%)

2. You are given a glass bottle and told that it is filled with water. You put it on a balance and find that it has a mass of 150.0 grams. The bottle is then filled with alcohol, which has a specific gravity of 0.79. You put it back on a balance and find that it has a mass of 123.75 grams. What is the mass of the bottle? What is the volume of the bottle? (*Ans*: $m_B = 25.0$ grams, $V_B = 125$ cm^3)

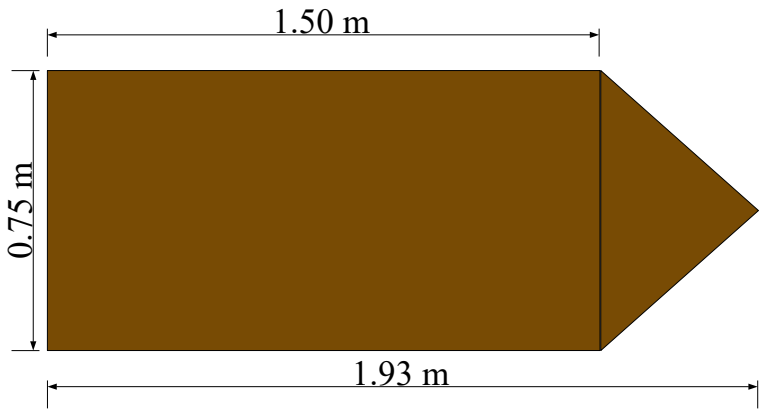
3. After college you decide to become a modern day treasure hunter. After locating an ancient statue that had fallen off of a ship in the Mediterranean Sea during the time of the Roman Empire, you dive down, hook up a line to it and then use a winch with a constant torque electric motor to pull it up. The motor is capable of providing a torque of $\tau_m = 3128.57$ N $\cdot$ m, and the radius of the spool of cable is 0.5 m. The statue is pulled up at a constant speed, and once on your boat, you find that it has a mass of 1041 kg. Note, the density of sea water is 1025 kg/m^3.

   (a) What is the volume of the statue? (*Ans*: 0.3927 m^3)

   (b) What is the density of the statue? (*Ans*: 2651 kg/m^3)

(c) Use the magic of the internet to find the density of marble. Could this statue be carved from marble? Look up the density of granite. Could the statue be made of granite? Is density alone enough to identify a material?

4. You are participating in a cardboard boat regatta. You design your boat so that it has a flat bottom but a pointy nose (see the diagram below for dimensions). You want to know how high the side walls of the boat should be, and you want at least 30.0 cm to be above the water line as a precaution against taking on water. The boat has a mass of 5.0 kg and you have a mass of 76.0 kg.



(a) What volume of water does the boat need to displace in order to stay afloat with you in it? (*Ans*: 0.081 m^3)

(b) Using the dimensions of the boat, how high up the sides will the water go? How high should you make the sides to build in your 30.0 cm buffer? (*Ans*: 6.30 cm, 36.30 cm)

5. A buoyant force will exist on any object submerged in a fluid. Air is a fluid, and all objects on the surface of the Earth are submerged in that fluid.

(a) Consider an object sitting on a scale. Write out the sum of forces on this object, including the buoyant force from the air, and derive an expression for the ratio of the apparent weight of the object to the actual weight of the object.

(b) The density of air is 1.29 kg/m^3. Find the density of an object that would have its actual weight reduced by 1% due to the buoyant force of the air. (*Ans*: 129 kg/m^3)

(c) Some wood has a density of about 300 kg/m^3. By what percentage is the weight of the this wood reduced due to the buoyant force of air? Do we need to worry about this effect when performing experiments? (*Ans*: 0.43%)

6. Water towers can be used to put pressure into the water system. A pump is used to push water up into the tank and then gravity creates pressure in the system. Assume that we have a cylindrical water tank with a radius of 5.0 m and the water is 5.0 m above the bottom of the tank. The tank is sitting on top of a 100.0 m tower.

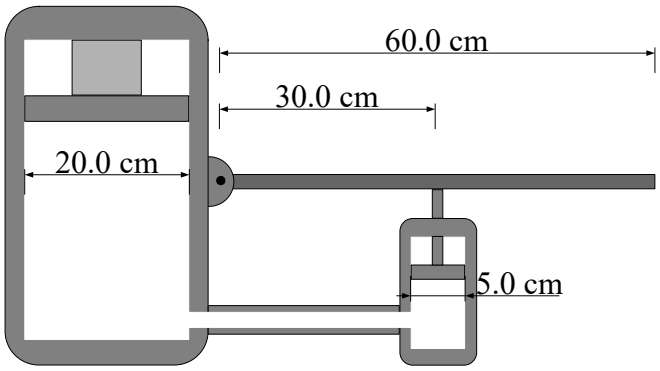
(a) What is the force of the water on the bottom of the tank? (*Ans*: 3.848×10^6 N)

(b) The pipe that feeds water into the system supplying nearby buildings, is 2.0 m below ground. What is the gauge pressure of the water at this point? (*Ans*: 1.0486×10^6 N/m^2)

(c) In a nearby two-story house, someone turns on a shower that is upstairs. The shower head is 5.5 m above the ground. Water flows in from the water main at the pressure found in the previous part of the problem, at a speed of 0.67 m/s through a 5 cm diameter pipe. As long as no other faucets are open, we can treat this as if there is a single pipe running from the water main to the shower head. If the pipe to the shower head has tapered down to a diameter of 2 cm, what is the speed at which the water exits that pipe? (*Ans*: 4.1875 m/s)

(d) What is the pressure at the shower head? (*Ans*: 9.6656×10^5 N/m^2)

7. Hydraulics make use of Pascal's principle to multiply an input force to a much larger output force. You decide to make a hydraulic press to test the compressive strength of materials. Your input cylinder has a diameter of 5.0 cm, and the output cylinder has a diameter of 20.0 cm. You use a lever to push down on a piston to increase the pressure in the hydraulic fluid. The lever is 60.0 cm long and the piston is pushed on by the middle of the lever. You prepare samples in such a way that when the lever is horizontal, the output piston is pressed against the sample, which in turn presses against the top of the press.



(a) You place a cube of concrete, 10.0 cm on a side, in the press (compressive strength $20{\times}10^6$ N/m^2). How much force will you need to apply to the end of the lever to cause the concrete to break? (*Ans*: 6250 N)

(b) You next place a cube of an unknown sample, still 10.0 cm on a side, in the press and find that it takes a force of 1.09×10^4 N on the lever to get the sample to crumble to pieces. What is the compressive strength of this unknown sample? (*Ans*: 3.488×10^7 N/m^2)

8. When you turn on a faucet, the stream of water gets narrower as it falls down towards the sink. This is particularly noticeable when you turn on the faucet of your kitchen sink at a lower flow rate.

(a) Derive an expression for the speed of the water as a function of the height fallen from the faucet, and the speed at which the water leaves the faucet, v_1.

(b) Derive an expression for the diameter of the stream of water as a function of the diameter of the faucet, d, the height fallen from the faucet, and the speed at which the water leaves the faucet, v_1.

(c) If the faucet is a height of 35 cm above the bottom of the kitchen sink, has a diameter of 0.5 inch, and the water leaves the faucet at a speed of 0.8 m/s, what is the speed of the water when it hits the sink and what is the diameter of the stream? (*Ans*: 0.686 cm)

9. A 5.0 m diameter, sealed tank is filled to a depth of 8.0 m. Air is pumped into the top of the tank so that the top surface of the water is under a gauge pressure P_t. A spigot on 0.5 m above the ground on the side of the tank is open so that water is flowing out into atmospheric pressure.

   (a) Assuming that the water at the top of the tank is essentially at rest, derive an expression for the speed, v_s at which the water will leave the spigot.

   (b) If $P_t = 3.14$ atm, what is the speed of the water? (*Ans*: 27.988 m/s)

10. **Numerical:** The Soviet Union sent landers to Venus in the late 1970s and early 1980s. They were able to send back pressure versus altitude date for Venus' atmosphere as follow (from Wikipedia):

Altitude (km)	**Pressure** (atm)
0	92.10
5	66.65
10	47.39
15	33.04
20	22.52
25	14.93
30	9.851
35	5.917
40	3.501
45	1.979
50	1.066
55	0.5314
60	0.2357
65	0.09765
70	0.03690
80	0.004760
90	0.0003763
100	0.00002663

   (a) Create a graph of the air pressure on Venus in Pascals (1 atm = 101325 Pa) versus altitude in meters.

   (b) Fit a second order polynomial to the data, making sure to display the equation and the R^2 value which describes the goodness of fit (the closer R^2 is to one, the better the fit). Also fit an exponential function to the data. Do either fit the data very well? Which one fits better?

   (c) If you are purely interested in describing how the pressure changes with altitude, you might resort to simply fitting higher order polynomial functions until you get a good fit to the data. Fit a sixth order polynomial to the data, disply the equation and R^2. Does this function fit the data well? Write down the function.

   (d) Using a second sheet in your spreadsheet, setup an altitude column that goes from 45,000 m to 55,000 m in increments of 50 m. Using the sixth order polynomial you found, calculate the pressure at each altitude. At what altitude is the air pressure equal to Earth's atmospheric pressure? (*Ans*: 50,332.5 m)

11 | Oscillations

1. We studying oscillations, we use the mass on a spring as the archetype. The force driving the oscillations is the spring force $F = -kx$, and this is then equal to the mass times the acceleration which yields

$$\frac{d^2x}{dt^2} = -\omega^2 x, \tag{11.1}$$

where $\omega = \sqrt{k/m}$. We then guess the solution and check via direct substitution into the above equation, i.e. given a function for $x(t)$, take two derivatives of it and see if it equals the right hand side.

   (a) Prove that $x(t) = A\cos(\omega t + \phi)$ is a solution to the above equation.

   (b) Prove that $x(t) = A\sin(\omega t + \phi)$ is a solution to the above equation.

   (c) Prove that $x(t) = Ae^{-i\omega t + \phi}$ is a solution to the above equation.

2. You attach a 2.00 kg mass to the right end a horizontal spring of constant $k = 25.13$ N/m. The left end of the spring is attached to a wall. The mass rests on a frictionless surface. You pull the mass back a distance of 10.0 cm and let go. You start your stopwatch when the mass is 5.00 cm to the right of the equilibrium position and heading towards the equilibrium.

   (a) What is the amplitude of the oscillations of this system?

   (b) What is the angular frequency of the system? Frequency? Period? (*Ans*: 3.5447 rad/s, 0.56416 Hz, 1.7726 s)

   (c) What is the phase angle for this system? (*Ans*: $\pi/3$ rad)

   (d) What is the equation that describes the position of the mass as a function of time? (i.e. Write out the equation by appropriately substituting the values you found in the previous parts.)

3. The position of a mass oscillating on a spring versus time is given by

$$x(t) = (19.0\text{ cm})\cos\left((8.38\text{ rad/s})t + 0.863\text{ rad}\right) \tag{11.2}$$

   (a) What is the amplitude of the oscillation?

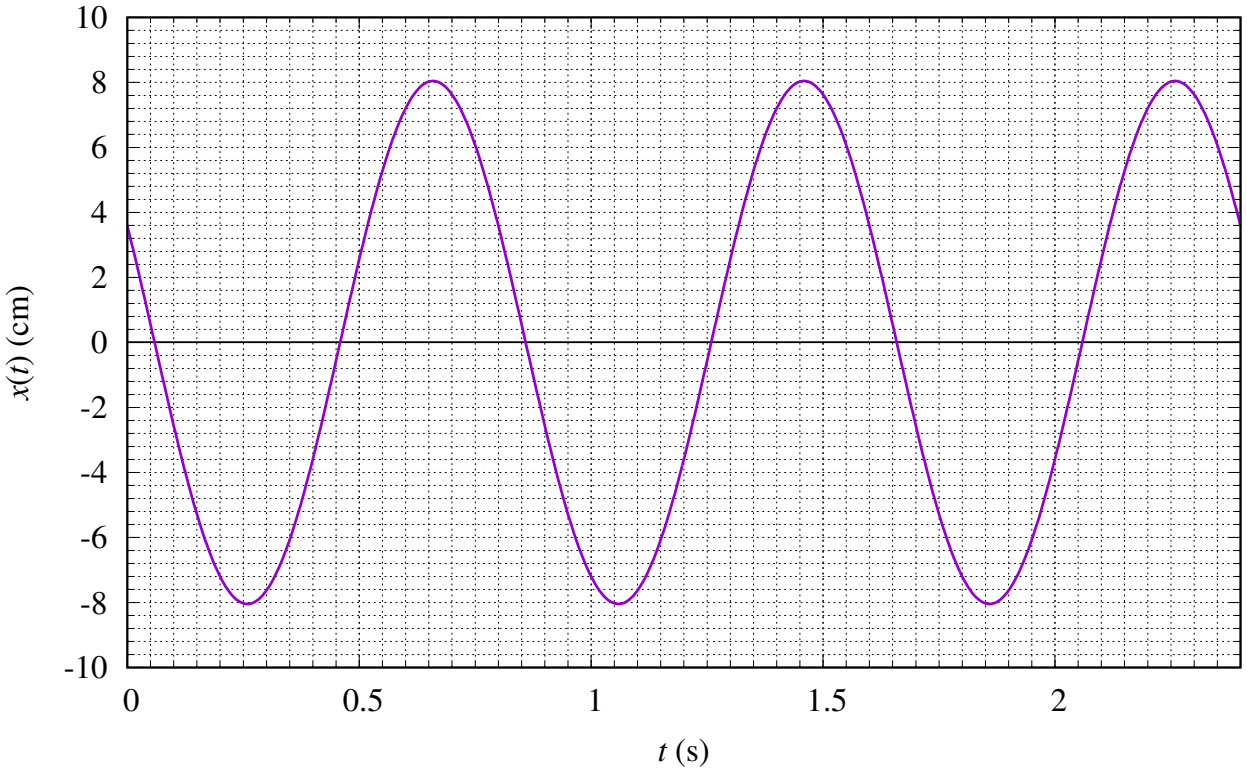
   (b) What is the angular frequency? Frequency? Period? (*Ans*: $f = 1.334$ Hz, $T = 0.7498$ s)

   (c) What is the phase angle?

   (d) What is the equation that describes the velocity as a function of time? What is the equation that describes the acceleration as a function of time?

(e) Create graphs of these equations for three full periods of the motion. This can be done in a spreadsheet by creating a time column with zero for the first entry, then have each subsequent time add 1/20th of the period to the previous time. Fill that column until you get to a time that is three times the period. Enter your equations for position, velocity, and acceleration in the next columns having the spreadsheet compute all the values for you. Make separate graphs position, velocity, and acceleration versus time as scatter plots with only smooth lines. Looking at the graphs, what is the velocity when the mass is at a maximum displacement? What is the acceleration?

4. The graph below shows the position of an oscillator as a function of time.



(a) Estimate the amplitude, period, frequency, angular frequency, and the phase angle of the oscillations. (*Ans*: $\omega = 5\pi/2$ rad/s, $\phi = 1.110$ rad)

(b) Write down the equation that would describe this motion.

(c) If the particle undergoing the oscillations has a mass of 140 g, what is the spring constant of the system? (*Ans*: 8.6359 N/m)

5. We have seen before that we can solve problems using forces or using energy and they both give equivalent descriptions of the motion. Now, let's examine this for the case of oscillations. You have a mass, m, resting on a frictionless surface, attached to a spring with constant k. You pull the mass back to a position $+A$, and then let go.

(a) What is the total energy in the oscillating system?

(b) Using conservation of mechanical energy, derive an expression for the velocity of the mass as a function of position x. **Note:** We have to use our knowledge of what will happen when

the mass is let go to get the proper sign of our function as the math will note give it to us automatically.

(c) Assume that the mass is 150 g, the spring constant is 94.7 N/m, and $A = 3.0$ cm. Write out the equation that represents the position as a function of time, and take its derivative to get the velocity as a function of time.

(d) Evaluate the position equation at $t = 2.4$ s, then evaluate your velocity as a function of position equation at that position, and your velocity as function of time equation at that time. Do they agree? Should they? (*Ans*: $v = 0.434$ m/s)

6. The text book and lecture videos go through a derivation of the equation of a simple pendulum using just force arguments, which require some potentially confusing steps. Since the pendulum bob moves along the arc of a circle, angular motion and torque is a more natural way of thinking about the problem. Start by assuming that we have a pendulum of length ℓ and mass m.

   (a) Draw a sketch of the pendulum displaced at an arbitrary angle to the right of vertical θ. On your sketch, draw the forces acting on the mass. **Note:** If you did this starting to the left of vertical, your angle would be negative, and we would have to make sure we account for that in our equations so we get a needed minus sign. This way, the minus sign should come from the torque.

   (b) Looking at the forces you have drawn, which force(s) will exert a torque on the pendulum bob about the point where the string is attached to a support? Write out the sum of torque equation.

   (c) What is the moment of inertia of the pendulum bob about the point where the string attaches to the support? How does the angular acceleration relate the angular position as a function of time? For example, if you were given and equation for $\theta(t)$, how would you get an equation for $\alpha(t)$? Use this information to setup an equation of motion for the pendulum in terms of θ. **Note:** For small angles $\sin\theta \approx \theta$.

   (d) Look at your equation. Does it look similar to the equation of motion for the mass on a spring? Use this to write down a solution to this equation. What is the angular frequency equal to for a pendulum? What is the frequency? What is the period?

7. After college, you become a high school science teacher. Your school is selected by NASA to design an experiment to take to the Moon on the first return trip by humans since the Apollo missions. You and your class decide that you want to measure the acceleration due to gravity on the moon using a specially designed pendulum that is completely encased, with a flap that holds the pendulum bob displaced by an angle of 5.00°. There is a switch on the outside that simultaneously releases the bob and turns on an infrared laser beam at the point where the bob passes through the vertical. A small computer then calculates the period by measuring the time between the beam being blocked.

   (a) You want to design the pendulum so that it has a period of exactly 1.000 s here on Earth. How long should you make the pendulum? (*Ans*: 0.2482 m)

   (b) Derive an expression for the acceleration due to gravity on the Moon, g_M in terms of period of the pendulum on the Moon, T_M, the period of the pendulum on the Earth T_E, and the acceleration due to gravity on the Earth g_E.

   (c) Once the experiment is run on the Moon, the Astronauts report back that the period was measured to be 2.458 ± 0.002 s. What is the acceleration due to gravity on the Moon? Estimate the error by calculating values for adding and subtracting the uncertainty in the period. (*Ans*: 1.622 ± 0.003 m/s^2)

(d) The mass of the Moon is $M_M = 7.342 \times 10^{22}$ kg, and its mean radius is $R_M = 1.7374 \times 10^6$ m. Calculate an expected value for the acceleration due to gravity on the Moon. Does your experimental value agree with this? (*Ans*: 1.622 m/s^2)

8. In the real world, all oscillations are damped oscillations due to friction forces or drag forces. Consider a pendulum constructed using a golf ball as the pendulum bob. As the ball swings, there will be a drag force from the air acting on the ball which will alter the period of the ball. The length of this pendulum is $\ell = 0.5$ m.

   (a) What is the expected period of this pendulum if we ignore drag? (*Ans*: 1.41923 s)

   (b) Let's say that we are able to measure the period very precisely and find that it is equal to 1.42012 s. What is the damping constant γ for the pendulum due to the drag force acting on it? **Hint:**
   $$\omega' = \sqrt{\omega^2 - \gamma^2} \tag{11.3}$$
   where ω' is the damped angular frequency, and ω is the undamped angular frequency. (*Ans*: 0.1567135 s^{-1})

   (c) The equation for the position of a damped pendulum as a function of time is
   $$\theta(t) = \underbrace{\theta_{\max} e^{-\gamma t}}_{A(t)} \cos(\omega' t). \tag{11.4}$$
   How long will it take for the amplitude of the pendulum to be half of the initial amplitude due to the damping? (*Ans*: 4.423 s)

   (d) How long will it take for the oscillations to stop (e.g. until they are $10^{-6}\theta_{\max}$)? (*Ans*: 88.158 s)

9. Automobile tires will always have small defects due to the manufacturing process. These defects often lead one part of the tire having a bit of extra mass compared to other parts of the tire. When tires are put onto a car they are first balanced by adding small weights to the rim of the tire on the point opposite that where they may be some extra mass. However, over time the tire can become out of balance again, leading to shaking of the vehicle that can be quite bad at particular speeds. A car's suspension represents a damped oscillator system. There is a spring and a shock absorber that serves to dampen the oscillations of the spring. The unbalanced mass of the tire causes there to be a driving force acting on this damped oscillator of the form $F_D = F_0 \cos(\omega t)$. The textbook and lecture videos point out that this leads to oscillations with an amplitude given by
   $$A_0 = \frac{F_0}{m\sqrt{\left(\omega^2 - \omega_0^2\right)^2 + b^2\omega^2/m^2}}. \tag{11.5}$$
   We want to figure out at what speed our car with an unbalanced tire will shake the most violently, which is to say at what speed will it have the largest A_0.

   (a) We can find the maxima and minima of a function by taking its derivative and setting it equal to zero.[1] Take the derivative of the above equation with respect to ω and then solve for ω to give an expression for the driving angular frequency that leads to the maximum amplitude.

[1] Recall that a derivative represents the slope of a function at a given point. The slope of the function goes to zero when it reaches either a maximum or minimum value. Taking the derivative, setting it equal to zero, and then solving for the variable we took the derivative with respect to then gives us all the values of variable where the function is a maximum or minimum. Generally, we would have to do more work to find out which is the case, but for this problem, there is only one point where the derivative equals zero.

(b) Assume that a car, $m_{car} = 1,120$ kg, has a suspension where the springs on each wheel have a spring constant of $k = 16,000$ N/m, and the shock absorbers have $b = 2000.0$ kg/s. Note that for springs in parallel, the effective spring constant is the sum of the individual spring constants. This means that for our car, we can treat each spring/shock system as supporting one quarter of the cars mass.[2] What is the driving angular frequency that leads to the maximum amplitude of oscillations? (*Ans*: 5.6243 rad/s)

(c) Because there is a bit of extra mass in one spot of the tire, there needs to be a bit of extra *centripetal force* at the edge of the tire there. This leads to an outward pointing force toward that spot at the hub of the wheel. This is the force that it driving the oscillations, however, only the vertical component of this force (i.e. the component parallel to the spring/shock system). This leads to a force of the form $F_V = F_C \sin(\omega_{rot} t)$. Draw a diagram that shows that this is the case. Compare this equation to the oscillating driving force equation given above. How does ω_{rot} compare to the driving angular frequency ω?

(d) If the wheel has a radius of 43.0 cm, at what speed will the car vibrate the most? (*Ans*: 2.418 m/s)

10. **Numerical:** When deriving the equation of motion for a pendulum, we have to make the approximation that $\sin\theta \approx \theta$. This approximation comes from the series expansion of the sine function around $\theta = 0$

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \ldots \tag{11.6}$$

When θ is a small number (i.e. less than one) raising it to a positive power will make it even smaller. If we are then dividing that even smaller number by the factorial of the positive power, it starts to get pretty tiny. So, the terms in our series expansion become less and less important. This lets us approximate the value of $\sin\theta$ as simply θ for small angles. Let's see how much of an effect this approximation can have.

(a) Start by creating a spreadsheet where the first column is the angle in degrees from 1 to 45 in one degree incriments. In the second column, store the angle converted to radians. In the third column, calculate the sine of the angle (you can point to the column where this is already in radians to make things easier). Lastly, in the fourth column compute the percent difference between θ in radians and $\sin\theta$,

$$\%\ \text{diff} = \frac{|\theta - \sin\theta|}{\sin\theta} \times 100\%. \tag{11.7}$$

(b) Create a scatter plot of the percent difference versus the angle in degrees. Choose the option to have smooth lines and the points on the plot. Using the graph, find the angles at which the small angle approximation is 1%, 5%, and 10% off the actual value of the sine function. Based on this, at what starting angle θ_{max} do you think a period of the pendulum would be significantly different than the predicted value based on the small angle approximation?

(c) **BONUS: This part of the problem can earn you 20 bonus points on this assignment!** Simply looking at where the small angle approximation breaks down gives you an idea of how small you should keep the amplitude, however, it doesn't actually tell us the effect on the period. For that, we need to simulate the motion of a pendulum and determine the exact period for different amplitudes. I have already collected this data for you, simply download the exactPen.txt file provided. To analyze this file, we have to use a somewhat more advanced

[2] This is an approximation since the mass of a car is not equally distributed.

piece of software than the typical spreadsheet, AlphaPlot.[3] Clicking on AlphaPlot will take you to the download page for the software. Go there, download the appropriate version for your computer's operating system, and then install the software.[4] If you are using Windows, it may complain about trying to install the software, you can safely ignore any of these warnings. AlphaPlot is an open-source program, meaning you can download and view the actual source code of the program to make sure there is nothing malicious. Once installed, launch the software and follow the steps below:

i. Start by clicking the 'x' on the Table 1 window and then select to delete it. This isn't completely necessary, but it does get the empty window out of the way.

ii. Click the file menu and select the "Import ASCII" option. In the pop-up window, navigate to where you downloaded exactPen.txt and click on the file once to select it. In the bottom portion of the pop-up window, in the "Separator" drop-down menu, select "SPACE". Once done, click the "Open" button on the right hand side of the pop-up window, roughly vertically centered.

iii. You should now have a table with 11 columns. Use your mouse to resize the Table 1 window horizontally, so that you can see all of the columns. The first column is the time, the other columns are then the angular position of the pendulum at the times listed in the first column. Each position column is for a different amplitude: 1°, 5°, 10°, 15°, 20°, 25°, 30°, 35°, 40°, and 45°, respectively. Click on the column header for the time then, while holding down shift, click on the header for the last column. This should highlight all of the table for you.

iv. On the left hand side of the program window is a stack of icons. The top one should look like a scatter plot. Click on that icon and then select the very first option that simply says "Scatter". A new window should appear with all of the data beautifully plotted in a rainbow of colors.

v. Again, resize this window so that you can see it a bit better (this isn't completely necessary). Then, right-click anywhere in the plot window (probably control-click for Mac users) and mouse over the "Analysis" entry, then select the Fit Wizard option. In the bottom half of the Fit Wizard window, type in the name "Oscillator No Phase", and for parameters enter "A, T" (exclude the quotes around these). In the big white space at the bottom type, without the quotes, "A*cos(2*pi*x/T)" to define the function we want to fit to the data. Here A represents the amplitude of the oscillations, and T is the period. Click the "Save" button in case you need to use this function again later (it will be under the "User defined" category), then click the "Fit»" button.

vi. In this next area, the "Curve" menu tells us which data we will be fitting to, we can leave it on the default selection for now. Near the middle, you should see the parameters that we defined listed, along with their starting values. Click on the number next to A and enter 0.017 (roughly 1° converted to radians), and the click on the number next to T and enter 1.4 (roughly the expected period for a 0.5 m long simple pendulum). Then, click on the "Fit" button at the bottom. The values should automagically change to their best-fitting

[3] AlphaPlot was selected because, while it does how much more advanced functionality than spreadsheets, it has a mostly point- and-click style interface.

[4] On the off chance that any of you is using a Linux based operating system (Ubuntu, Mint, Fedora, OpenSUSE, Arch, Manjaro, etc.) try searching for AlphaPlot in your distributions package manager. It is an open source program, so it may be available there (it is in the AUR). If not, it is available as a Flatpak, which will likely be the easiest way to get it installed and working (this is the version I went with).

values. Record the period, or copy and past it into a spreadsheet, for further analysis later, making sure to note that this is for an amplitude of 1°.

vii. Click the curve drop down menu and select the second one listed, corresponding to an amplitude of 5°, and then simply click the "Fit" button again. Record the period for this amplitude.

viii. Repeat this until you have the period for all of the data in the graph. Make sure to note the angle each one is associated with, given by the number at the end of the curve name. Then click the "Close" button to exit the Fit Wizard.

ix. Enter this data into a spreadsheet with the first column as the angle, the second as the period you found from AlphaPlot. Compute the expected period for a simple pendulum with a length of 0.5 m using

$$T = 2\pi\sqrt{\frac{\ell}{g}}. \tag{11.8}$$

In a third column, find the percent difference of the period you found from AlphaPlot, and the expected period if the pendulum behaved as a simple-harmonic oscillator. Create a scatter plot of this percent diffence versus the amplitude, selecting the points and smooth lines option. From your graph, find where the period will differ by 1%, 5%, and 10%. Based on your findings, what's the largest angle that you would feel comfortable saying that a pendulum behaves like a simple-harmonic oscillator?

12 | Waves & Sound

1. You attach a piece of string that is 3.0 m long and has a mass of 1.98 grams to a table, and then run the string over a small amplitude mechanical oscillator, and then over a pulley. On the end of the string that runs over the pulley you tie a mass hanger.

   (a) If the distance between the mechanical oscillator and the pulley is 2.25 m and you have 300 grams on the mass hanger, what frequency should you set your mechanical oscillator to in order to see one antinode on the string? Two antinodes? Five antinodes? (*Ans*: 14.83 Hz, 29.66 Hz, 74.16 Hz)

   (b) If you set the oscillator to 100.0 Hz, how much mass would you need on the hanger to see one antinode? Two antinodes? Five antinodes? (*Ans*: 13.64 kg, 3.409 kg, 0.5455 kg)

   (c) With the oscillator set to 100.0 Hz, and the mass on the hanger set to 500.0 grams, how much closer to the pulley would you have to move the oscillator in order to see a standing wave pattern? How many antinodes would this pattern have? (*Ans*: 0.0959 m closer)

2. Submarines can use Sonar to map out their surroundings. This works like echolocation, a sound (which is a longitudinal wave) is emitted, usually at a high frequency, and then based on the time it takes for the sound to return, distances can be determined. A "ping" is sent out, and an echo is heard 1.42 s later. How far away is the object? **Note:** $B = 2.34 \times 10^9$ N/m^2, and $\rho = 1025$ kg/m^3 for sea water. (*Ans*: 1073 m)

3. We are told that the equation that describes the displacement of a medium at a point x as a function of time as a wave travels through the medium is

$$D(x,t) = A\sin(kx - \omega t) \tag{12.1}$$

   where k here is the wavenumber[1], and ω is still the angular frequency. This equation should be the solution to the wave equation

$$\frac{\partial^2 D}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 D}{\partial t^2} \tag{12.2}$$

   where the $\partial/\partial x$ (or t) denotes the *partial derivative*. The partial derivative is basically the same as a regular (or total) derivative, but we hold all other variables constant. For example,

$$f(x,y,z) = x^2yz^3, \quad \longrightarrow \quad \frac{\partial f}{\partial x} = 2xyz^3, \quad \frac{\partial f}{\partial y} = x^2z^3, \quad \frac{\partial f}{\partial z} = 3x^2yz^2. \tag{12.3}$$

[1] I realize this can be confusing as we were just using k to be the spring constant, and soon we will use k to be Boltzmann's constant. Unfortunately, the text book author chose not to use other symbols here. For example, my preference would have been to use k for the spring constant, the Greek letter κ (kappa) for the wavenumber, and k_B to distinguish Boltzmann's constant.

Take the implied partial derivatives of the $D(x,t)$ equation, and prove that it is a solution to the wave equation.

4. You are standing next to a small garden pond and throw a small stone into the middle. You notice that the ripples travel outwards in circles, and that their amplitude decreases as they travel further from the point where the stone went in. The *intensity* of a wave is the power per unit "area" (this could be an actual surface area for 3D waves, or it could be a perimeter for 2D waves)

$$I = \frac{\overline{P}}{S}, \quad \longrightarrow \quad \overline{P} = IS = 2\pi^2 v\rho f^2 A^2 S. \tag{12.4}$$

As long as there is no (or negligible) loses, then the total power begin radiated by a ripple should be constant.

   (a) What is the "area," S, for a circular ripple in water?

   (b) Using this S and equation (12.4), derive an expression that relates the amplitude, A_1, at a distance from the source (where the stone went in), R_1, to the amplitude, A_2 at another distance R_2 from the source.

   (c) If the amplitude of the ripples is 0.1 cm at a distance of 6.00 m from the source, what is their amplitude at a distance of 2.00 m from the source? (*Ans*: 0.173 cm)

5. The principle of superposition tells us that the net displacement of a medium when multiple waves are traveling in it is equal to the sum of the displacements from each wave. Sound is, of course, just an example of a longitudinal wave and if you have stereo speakers, you have two wave sources. Music is going to be the superposition of a bunch of different wavelengths and frequencies, but let's assume for a moment that we have two speakers emitting equal amplitude waves of a pure tone (i.e. same frequency and wavelength). These waves can be represented as

$$D_1(r_1,t) = A\sin(kr_1 - \omega t), \quad \text{and} \quad D_2(r_2,t) = A\sin(kr_2 - \omega t + \phi), \tag{12.5}$$

where r_1 and r_2 are the straight-line distances from the speaker to the location of an observer, and ϕ represents the phase difference between the speakers.

   (a) Shows that the net displacement of the air molecules is

$$D(r_1,r_2,t) = D_1 + D_2 = 2A\cos\left(\frac{k(r_1-r_2)}{2} - \frac{\phi}{2}\right)\sin\left(\frac{k(r_1+r_2)}{2} - \omega t + \frac{\phi}{2}\right). \tag{12.6}$$

   The following trigonometric identity will be needed

$$\sin\theta_1 + \sin\theta_2 = 2\sin\left[(\theta_1+\theta_2)/2\right]\cos\left[(\theta_1-\theta_2)/2\right].$$

   (b) The sine portion of equation (12.6) is the part that oscillates with time. The cosine part varies with position in space, and determines the maximum amplitude at different places. Looking at the cosine function, what would the argument of the cosine function need to be equal to in order to have maximum constructive interference (net displacement of $2A$)? What value would be needed to have maximum destructive interference (net displacement of 0)? Is there a single value or multiple values it could equal?

(c) Assume you have two speakers sitting a distance of 1.5 m apart on a table, emitting the same pure tone (512 Hz) exactly in phase (i.e. $\phi = 0$). You place a microphone 2.0 m in front the table at the midpoint between the speakers. How far to the right would you need to move the microphone so that a minimum in the sound level is recorded?[2] How far to get a maximum sound level? (*Ans*: 0.460 m, 1.053 m)

(d) What would happen if the two speakers are exactly out of phase?

6. In lab there is a pipe with one end sticking out from a large curtain. At the end of the pipe that you can see, there is a small speaker connected to a frequency generator so that you can have it produce pure tones of any frequency. You slowly sweep through the frequencies, noting that you first get resonance (i.e. amplification of the sound) at 71.458 Hz, which is then followed by resonance at 214.375 Hz, 357.292 Hz, and 500.208 Hz, but no resonance at any frequency between these.

   (a) Is this an open or a closed pipe?

   (b) What is the length of the pipe? (*Ans*: 1.200 m)

7. A guitar has 6 strings, all of which have the same length, 0.6447 m, but have different linear mass densities. The tension of each string can be adjusted to tune the guitar so that the fundamental frequency produced by the string is set to the desired frequency. The strings have the following characteristics.

String	**Linear Mass Density, μ** (kg/m)	**Tuned Frequency** (Hz)
Low E	5.78×10^{-3}	82.41
A	3.50×10^{-3}	110.0
D	1.62×10^{-3}	146.83
G	1.03×10^{-3}	196.0
B	4.77×10^{-4}	246.94
High E	3.09×10^{-4}	329.63

   (a) What tension must the low E string be under in order to be in tune? (*Ans*: 65.262 N)

   (b) Assuming its properly tuned, when you push you finger on the string 10.0 cm down (i.e. the lenght of the string is reduced by 10.0 cm), what frequency will the low E string produce now? (*Ans*: 97.539 Hz)

   (c) When you pluck the D string, your tuner tells you it is producing a frequency of 132.7 Hz, what is the tension in that string? (*Ans*: 47.428 N)

   (d) Will you have to increase or decrease the tension to bring it into tune?

8. You buy a Bluetooth speaker that has an average power output at full-volume of 16 Watts.

   (a) What is the decibel level of this speaker at full-volume if you are standing 1.00 m away? (*Ans*: 121 dB)

[2] This wouldn't be dead silence since the amplitude of each wave depends on the distance from the speaker that emits it. If the distances from the speakers to the point are different, this means the amplitudes of the soundwaves won't be exactly the same so you won't get perfect cancellation.

(b) Assuming that 50% volume is half as loud as 100% volume, what is the power output of the speaker at 50% volume? (*Ans*: 1.6 Watts)

(c) If the speaker is at full-volume, how far away would you need to be for the sound to be half as loud as it is at 1.00 m away? (*Ans*: 3.162 m)

9. Going back to problem 2, if the submarine ping was sent out with a frequency of 40 kHz, the submarine is travelling at 5 m/s in the direction of the detected object, and the frequency of the echo was recorded at 40.454 kHz. How fast and in what direction (towards or away) is the object that was detected moving? (*Ans*: 7.069 m/s)

10. **Numerical:** In the textbook and lecture videos you are told that the speed of sound in air depends on temperature. This should make some sense because the density of air changes with temperature. The *ideal gas law*, which we will explore in more detail later, tells us that

$$PV = nRT \tag{12.7}$$

where P is the pressure, V is the volume, T is the temperature, n is the number of moles of gas, and R is the gas constant. We know that density is mass over volume, so we get

$$\rho = \frac{m}{V} = \frac{mP}{nRT} = \frac{\mu P}{RT} \tag{12.8}$$

where we have substituted $V = nRT/P$, and noted that the molar mass, μ, is the mass divided by the number moles, $\mu = m/n$. We learned that for longitudinal waves, the speed of the wave in a fluid (gas or liquid) is

$$v = \sqrt{\frac{B}{\rho}}, \tag{12.9}$$

and since we know how the density changes with temperature, if we knew the bulk modulus of air we could find a relationship that would tell us how the speed of sound changes with temperature. Turns out that the bulk modulus of gases is related to the pressure of the gas, and for air happens to be $B_{\text{air}} = 1.4P$.[3] So, we get

$$v = \sqrt{\frac{1.4PRT}{\mu P}} \quad \longrightarrow \quad v = \sqrt{\frac{1.4RT}{\mu}}. \tag{12.10}$$

Note that in this equation we need to input the temperature in Kelvin (absolute temperature units). The value of molar mass for air is, $\mu = 0.02897$ kg/mol, and the value of the gas constant is $R = 8.314$ J/K mol. So, this tells us the speed of sound in air at any temperature T. So, why does the book tell you that the relation is

$$v \approx (331 + 0.6T) \text{ m/s} \tag{12.11}$$

where T is in Celsius? To answer this question do the following:

[3]The factor out front here is the heat capacity ratio, $\gamma = C_{\text{P}}/C_{\text{V}}$, where C_{P} is the specific heat at constant pressure, and C_{V} is the specific heat at constant volume. The specific bulk modulus that we are calculating is then called the isoentropic (constant entropy) bulk modulus.

(a) Create a spreadsheet where the first column is the temperature in Kelvin, starting at absolute zero (i.e. just zero) and going to 400. In the second column, convert this temperature to Celsius by subtracting 273.[4] In the third column, compute the speed of sound in air at the particular temperature using equation (12.10), make sure to reference the Kelvin temperature in the formula, we will use the Celsius temperature to make a graph later.

(b) Create a scatter graph of the speed of sound versus the Kelvin temperature for a range of 0 K to 400 K, choosing the smooth line only option. Is your graph linear? Does any part of it look linear?

(c) Create a scatter graph of the speed of sound versus the Celsius temperature in the range of -40°C to 40°C (this is -40°F to 104°F) which covers the majority of the range of temperatures experienced on Earth. Again, choose the smooth line only option. Does your graph look linear?

(d) Fit a linear trend line to the second graph, making sure to display the equation and the R^2 value. How does your trend line's equation compare to equation (12.11)? **Hint:** It should match pretty much exactly, showing you where that particular equation comes from.

[4] 1 Kelvin is the same size as 1 Celsius degree, so we just need to adjust for the zero-point offset.

13 | Atomic Theory – Temperature – & Gas Laws

1. You find an old mercury thermometer in chemistry lab, but the markings have faded. You make a mark and label it with 21.7°C, the current temperature in the lab according to a digital thermometer on the wall. You then put the thermometer into a beaker of boiling water, wait for the column of mercury to stop rising, and then make another mark. You measure 15.0 cm between your two marks, and note that the length of the thermometer which used to have markings is 30 cm long.

   (a) How many Celsius degrees are there per centimeter of length for this thermometer? If you made marks every millimeter, what would the accuracy of this thermometer be? (*Ans*: 5.22 C°/cm, 0.522 C°)

   (b) You notice that the old scale went 5 cm below your first mark, and 10 cm beyond your second mark. What was the lowest and highest temperature you could read with this thermometer? (*Ans*: -4.4°C, 152.2°C)

   (c) You place the thermometer into a solution undergoing an exothermic reaction. You make a mark and find that the level of the mercury is 5.1 cm above your first mark. What is the temperature of the solution? (*Ans*: 48.322°C)

2. You probably know that when materials heat up, they tend to expand. The amount of expansion depends on the change in temperature, and the amount of material you have. This should make some sense, if for a given change in temperature, two atoms in a solid spread apart by a certain amount, then that effect is going to be additive based on how many atoms you have in total. So, the equation that gives you the new length of a thin rod of material is

$$\ell = \ell_0(1 + \alpha\Delta T) \tag{13.1}$$

   where α is the coefficient of linear expansion and depends on the particular material.

   (a) You have a cube of material with sides of length ℓ_0 at room temperature (20°C). Write out an expression for the volume of the cube after a temperature change of ΔT in terms of its volume at room temperature, the coefficient of linear expansion, and the change in temperature. Assume that the cube expands isotropically (i.e. the same in all directions).

   (b) The coefficient of linear expansion is generally a very small number, $\sim 10^{-5}$ for many materials. This means that you would need a temperature change of $\sim 100,000$ C° to have $\alpha\Delta T \approx 1$, or in other words, $\alpha\Delta T \ll 1$ for the vast majority of situations, and when it's not the material has probably vaporized by then. Assume that $\alpha = 10^{-5}$ $(\text{C}°)^{-1}$ and $\Delta T = 100$ C°. How does $\alpha\Delta T$ compare to $(\alpha\Delta T)^2$ and $(\alpha\Delta T)^3$? Looking at your expression for the volume from part (a), do you think there are terms that can be safely ignored?

(c) Use your expression from part (a) to write down the change in volume in terms of the initial volume, the change in temperature and the coefficient of linear expansion. Get rid of any terms that you can safely ignore based on your findings from part (b). Compare your expression to

$$\Delta V = \beta V_0 \Delta T \tag{13.2}$$

where β is the coefficient of volume expansion. What should β be approximately equal to in terms of α?

3. Venus is an extreme planet. The surface pressure on Venus is about 93 atm, and the temperature is about 453°C. If we send a sphere of aluminum ($\beta = 2.5 \times 10^{-5}$ (C°)$^{-1}$, $B = 7.0 \times 10^{10}$ N/m^2) to Venus, the increase in temperature would tend to increase its volume, while the increase in pressure would tend to decrease its volume. Any change in volume of an object will lead to a change in density.

   (a) A sphere of aluminum with a diameter of 0.15 m at room temperature (20°C) is sent to Venus. What is its volume of the surface of Venus? Does it expand or compress overall? (*Ans*: 1.889×10^{-5} m^3)

   (b) What is the fractional change in density of the sphere? (*Ans*: -0.01058 or -1.058%)

4. Absolute zero comes from an extrapolation of Charles' Law, where he found that the volume of a gas is directly proportional to its temperature, i.e. if you plot the volume of a gas as a function of temperature you get a straightline. When this line is extrapolated backwards, it crosses the temperature axis at −273.15°C. We just saw that the change in volume of a substance should be

$$\Delta V = \beta V_0 \Delta T, \quad \text{meaning} \quad V = V_0(1 + \beta \Delta T). \tag{13.3}$$

   (a) What has to be true for the volume to go to zero?

   (b) Air has a coefficient of volume expansion of $\beta = 3.4 \times 10^{-3}$ (C°)$^{-1}$ at room temperature. Assuming that β is constant, what temperature corresponds to zero volume? (*Ans*: −274.12°C)

   (c) Why do you think your answer doesn't exactly agree with −273.15°C?

5. A helium tank for filling party balloons has a volume of 0.164 m^3 and is filled with 17.55 moles of helium at room temperature (293 K).

   (a) What is the mass of the helium in the tank? (*Ans*: 70.2 g)

   (b) What is the pressure in the tank? (*Ans*: 2.6068×10^5 Pa)

   (c) Is the buoyant force on the tank greater than the weight of the helium?

6. Most modern cars have tire pressure sensors in the wheels to inform the driver if a tire has low pressure. However, gases will expand and contract with temperature. You filled your tires with air on a nice warm day with a temperature of 25°C, to a pressure of 32 PSI (221 kPa). Your tire pressure sensors will trip if the pressure drops to 28 PSI (193 kPa). How cold would it need to get for the temperature change to be enough to trip your pressure sensors? (*Ans*: -12.77°C)

7. Consider 1.00 m^3 of air (which can be treated as an ideal gas) at 293 K and 1.00 atm.

   (a) How many molecules of air is in that 1.00 m^3? (*Ans*: 2.5048×10^{25})

(b) If we model the gas molecules as spheres with diameters of 3.00×10^{-10} m, how much volume would one molecule occupy? (*Ans*: 1.414×10^{-29} m^3)

(c) How much volume is occupied by all the molecules in the 1.00 m^3 if they are packed as close together as possible? (*Ans*: 3.541×10^{-4} m^3)

(d) How much pressure would you have to put the air under to get it packed into that volume? (*Ans*: 2.860×10^8 Pa)

8. (a) Use the ideal gas law to show that the root mean square speed of a gas can be written as

$$v_{\text{rms}} = \sqrt{\frac{3P}{\rho}}, \tag{13.4}$$

where P is the pressure the ρ is the density.

(b) For an ideal gas at temperature T, show that

$$\frac{dv_{\text{rms}}}{dT} = \frac{1}{2}\frac{v_{\text{rms}}}{T} \tag{13.5}$$

and that if we approximate $\Delta v_{\text{rms}} \approx (dv_{\text{rms}}/dT)\Delta T$, then

$$\frac{\Delta v_{\text{rms}}}{v_{\text{rms}}} \approx \frac{1}{2}\frac{\Delta T}{T}. \tag{13.6}$$

(c) The average daytime high temperature in January in Pittsburg, KS is 6.7°C, while in July it is 31.7°. Using the expressions from part (b), what is the fractional change in the root mean square speed of the air molecules between these months? (*Ans*: 1.866)

9. The average surface temperature on Venus is 726 K, and its atmosphere is ∼96% carbon dioxide. The surface pressure is ∼93 atm.

(a) Approximating the atmosphere as pure carbon dioxide, what is the root mean square speed of the molecules? What is the average speed of the molecules? What is the most probable speed of the molecules? (*Ans*: 641.31 m/s, 590.85 m/s, 523.63 m/s)

(b) How many molecules are contained in a volume of 1.00 m^3? (*Ans*: 9.4055×10^{26})

(c) What is the average distance between the molecules? (*Ans*: 1.021×10^{-9} m)

10. **Numerical:** In the textbook and lecture videos, we are told that the speeds of gas molecules follow a Maxwell distribution. But why is that? Consider a single dimension for a moment. In the x-direction, we would expect that any arbitrarily chosen molecule would have just as much of a chance to be going in the positive direction as in the negative direction. For a large enough collection of molecules, there is likely going to be one going in the positive direction with a speed v, and one going in the negative direction with speed v, so we would expect the average *velocity* in the x-direction to be zero. Since there is nothing special about the direction that we arbitrarily call x, we would expect the same in the y and z-directions as well. We can then calculate the expected spread in velocities noting that in one-dimension, the average kinetic energy per particle should be

$$\frac{1}{2}m\overline{v^2} = \frac{1}{2}kT^{[1]} \quad \longrightarrow \quad v_{\text{rms}} = \sqrt{\frac{kT}{m}}. \tag{13.7}$$

[1]The 3 we are familiar with comes from the fact that we get a kT/2 in the average energy for each degree of freedom for the molecules/atoms. A 3D ideal gas can move in 3-dimensions, therefore 3 degrees of freedom.

This is useful because it can tell us about the spread in our velocity. The *variance* of a quantity (the standard deviation squared) is given by

$$\sigma^2 = \overline{v}^2 + \overline{v^2}. \tag{13.8}$$

Since we reasoned that the mean value should be zero, the square of that is still zero. So, the standard deviation of the velocities of particles in one-dimension is just the root mean square speed in that direction,

$$\sigma = \sqrt{\frac{kT}{m}}. \tag{13.9}$$

Now, let's see how having three Gaussian (normal) distributions for the components of velocity leads to a Maxwell distribution for the speeds.

(a) Download the spreadsheet provided on Canvas. This should have nine labeled columns, the first one just being a particle number to help you more easily fill in the columns next to it.

(b) The next three columns labeled P_x, P_y, and P_z are for the probability of finding a particular velocity in the x, y, and z-directions, respectively. Next to particle #1, enter the equation "=rand()" in the cell. This function returns a random number between 0 and 1 with each number in that range have an equal probability. Mouse over the bottom right corner of the cell, and drag this across below the P_y and P_z columns, then double click the bottom right corner to fill all the cells below. **Note:** The numbers in the cells from these random numbers will keep changing as you work. This is fine and will not affect your outcome.

(c) Off to the right of the labeled columns you should find some cells with labels for different numbers. Next to the one labeled T, enter a temperature of 293 K (omit the units when entering). The cell next to sigma should then fill with the expected value based on equation (13.9).

(d) In the first empty cell in the v_x column, enter "=norm.inv(B2, 0.0, M4)" which will give you a velocity drawn from a Gaussian distribution with a mean of 0.0 and a standard deviation of σ that has the probability equal to the value in B2. Mouse over the bottom right corner of the cell, and drag the formula over to the v_y and v_z columns, then double click to fill all the columns below that.

(e) In the next column, compute the square of the velocity described by the components in the preceeding columns.

(f) In the last column, compute the magnitude of the velocity, i.e. the speed, by taking the square root of the velocity squared column.

(g) Next to their labels, compute the average velocity and the root mean square velocity from the data in your table. Note that the cell range of interest is 2 to 10001, so it might be easier to manually type the cell range than to highlight it with a mouse. Using the approprate equations,

(h) The average speed and the root mean square speed are related to the temperature by

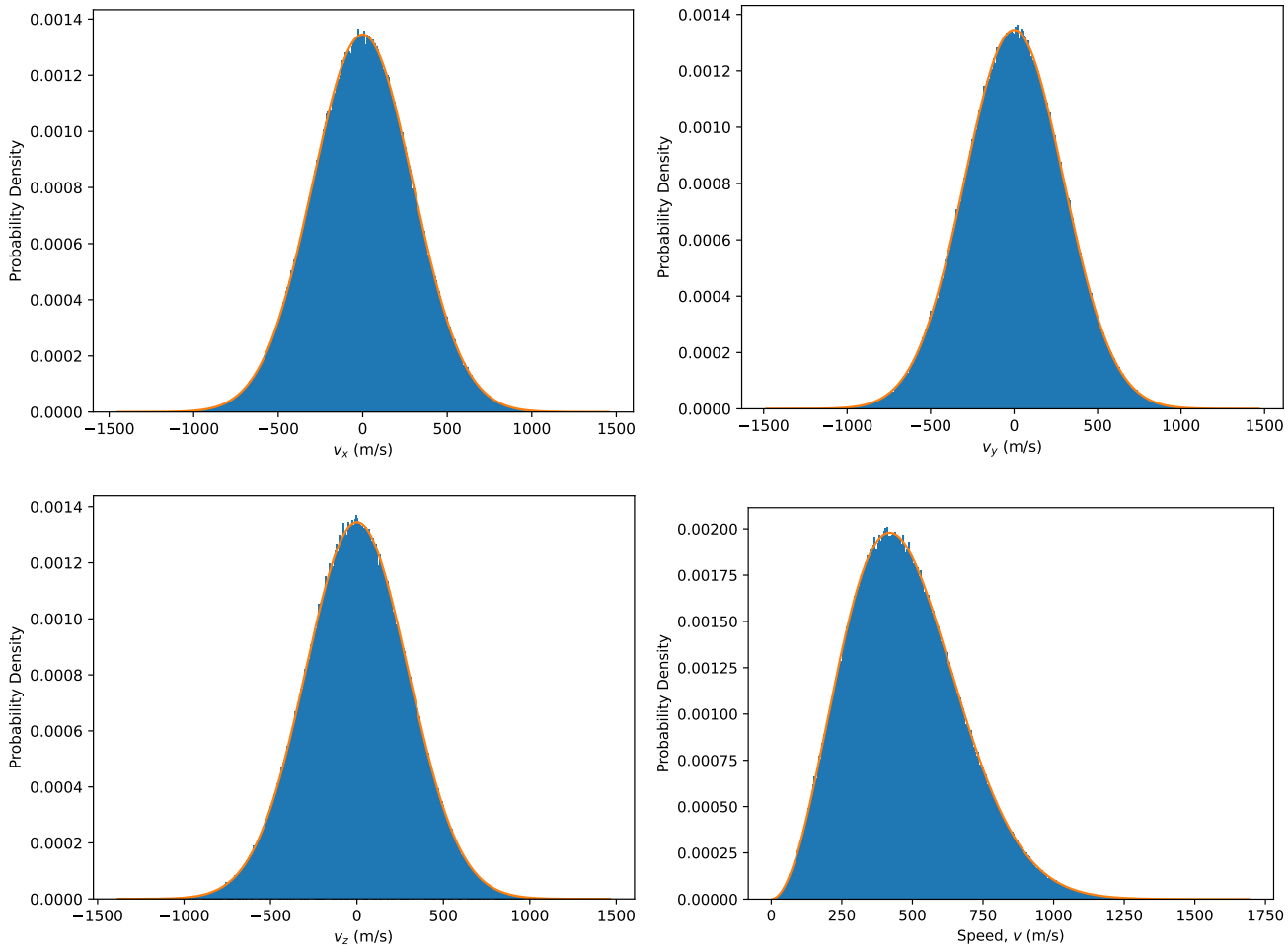
$$\overline{v} = \sqrt{\frac{8kT}{\pi m}}, \quad \text{and} \quad v_{\text{rms}} = \sqrt{\frac{3kT}{m}}, \tag{13.10}$$

respectively. Use the equations to find the temperature of the gas from the speeds you found in part (g). Do they reasonably agree with the input temperature of 293 K? What is their percent difference?

(i) Look at Sheet 2. There you will find a histogram of the x, y, and z-components of the velocity plotted against a theoretical Guassian distribution with a mean of zero and a standard deviation of $\sqrt{kT/m}$. Do the lines for the components follow closely to the theoretical line?

(j) Look at Sheet 3. There you will find a histogram of the speeds of the particles plotted against a theoretical Maxwell distribution. Does the histogram line agree with the theoretical curve? You have now seen how three Gaussian distributed velocity components leads to a Maxwell distribution of speed.

The particle sample we looked at was only 10,000 particles and you should find that it's pretty much in line with our expectations. However, things look even better for larger numbers of particles. The plots below show what happens if we have 1,000,000 particles, with the blue bars showing the histogram and the orange line showing the theoretical expectation. Note, you should be able to zoom in on these plots as much as you want without losing sharpness, so make sure to take a close look. The temperature found from the average velocity here was 292.97 K and from the RMS velocity was 292.99 K, with and input of 293 K.

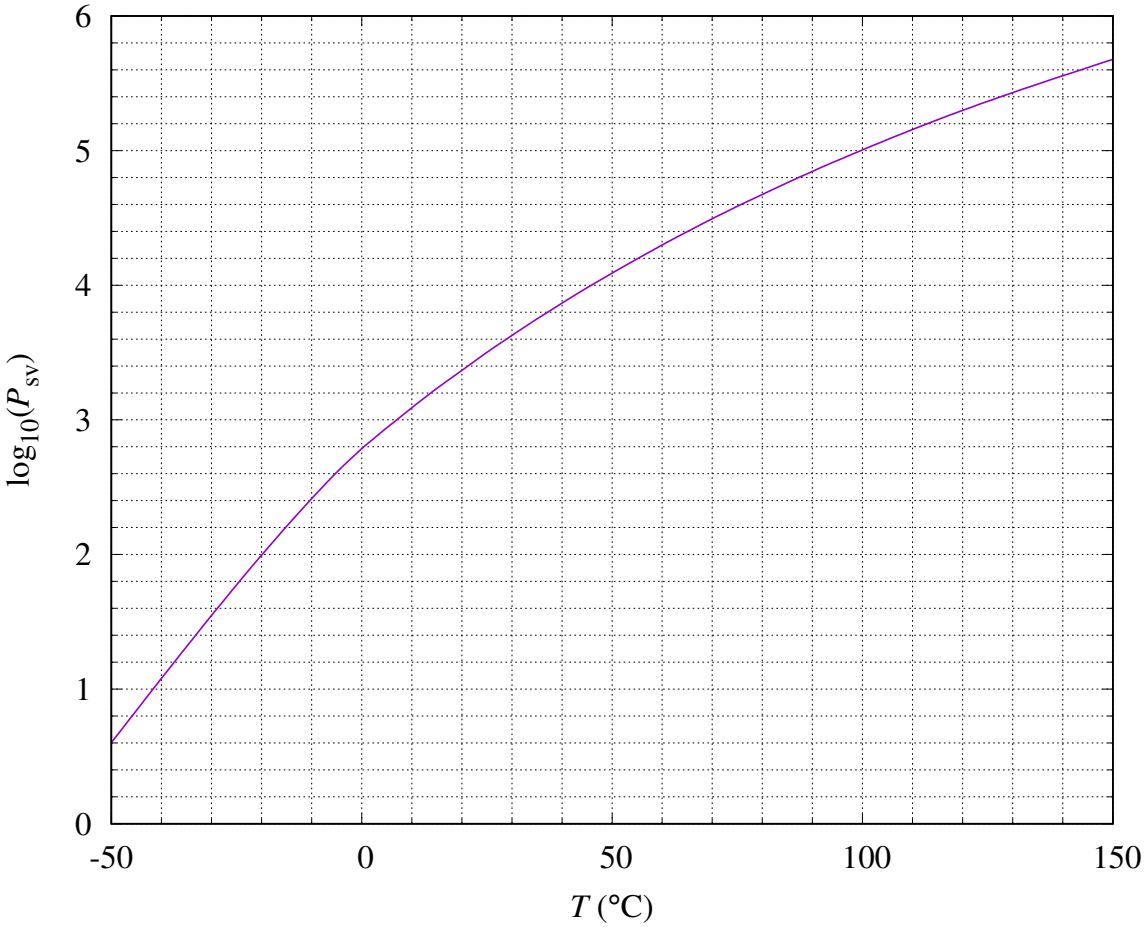


14 | Heat & the Laws of Thermodynamics

1. Pressure cookers work by raising the boiling point of water so that food can cook at a higher temperature. This is because the temperature of any substance doesn't change during a phase change.

   (a) Using the graph below, estimate the pressure inside the pressure cooker if you want your food to cook at 120°C. (*Ans*: 1.995×10^5 Pa)

   (b) The pressure cooker has a release value to prevent too much pressure from building up. This valve is held down with a spring that is slightly stretched by 2.0 cm. The circular valve has a diameter of 0.5 cm. What should the spring constant be so that the valve will just be held closed when the pressure cooker is at 120°C? (*Ans*: 96.41 N/m)



2. During the summer time, it often gets quite humid in addition to being quite hot. The air conditioning in your apartment is broken, so you have your windows open while waiting on your landlord to get around to fixing it. While you lie in bed trying to sleep in the very uncomfortable conditions, you wonder just how much water is in the air in your room. You remember that the weather man said the humidity was 71.5% during the day with a high temperature of 91°F (32.8°C). It has cooled down to 80°F (26.7°C).

   (a) What was the partial pressure of water vapor during the day? Use the graph above to find the appropriate saturated vapor pressure. (*Ans*: 3193.79 Pa)

   (b) Assuming the same water content and volume, what is the partial pressure with the temperature drop? (*Ans*: 3130.11 Pa)

   (c) What is the percent humidity at night? (*Ans*: 99%)

   (d) Assuming that your room is 3.66 m by 3.66 m with 2.74 m ceilings, what mass of water is contained in your room? (*Ans*: 0.8295 kg)

3. You are making a box of mac and cheese and the box tells you to bring 6 cups of water to a boil on your stove (1 cup = 236.6 ml).

   (a) Assuming that the water starts at about 10°C, how many calories of energy does it take to bring the water to a boil? How many joules is this? (*Ans*: 127764 cal, 534820 J)

   (b) Assuming that it take 5 minutes for the water to boil and that energy is delivered to the pot at a constant rate and that your electric coil burner stove is 75% efficient, what is the average power usage in Watts (J/s) of your stove? (*Ans*: 2376.98 W)

   (c) If electricity costs 13.72 cents per kWh (kilowatt hour), how much does it cost to bring the water to a boil for your mac and cheese? (*Ans*: 2.718 cents)

4. For a date, you decide you need to step up your cooking game from boxed mac and cheese to impress them. You purchase an immersion circulator so that you can sous vide some steak to medium rare at 130°F (54.4°C) with some head room to sear the steak after without over-cooking it.

   (a) If the immersion circulator is rated to 800 W and the water starts at 12°C, how long will it take to get the 8.0 l of water up to temperature? (*Ans*: 1774.9 s)

   (b) The specific heat of beef is about 1670 J/kg·C°. If the steak starts at refrigerator temperature, 4.4°C, how much heat energy will be transferred to the 8.0 ounce (226.8 g) steak to cook it to medium rare? What is this in calories? (*Ans*: 18937.8 J, 4524.08 cal)

5. You are making a cup of tea to go. You pour the boiling water into a double-walled glass travel mug (c = 0.20 cal/g·C° = 840 J/kg·C°). The inner glass wall is 300 g and you fill it to its 16 fluid ounce capacity (500.0 ml), and put the lid on. Assuming the glass starts at a temperature of 20.0°C, what is the equilibrium temperature? Assume that we can neglect heat losses to the environment. (*Ans*: 91.43°C)

6. You are trying to determine the identity of an unknown metal using calorimetry. The mass of the sample is 100.0 g. You place the sample in a beaker of boiling water while monitoring the temperature of the system (water + sample) with a thermometer. The system comes to an equilibrium temperature of 98.4°C. You place the sample in a 250 g of water (c_{H_2O} = 4.186 kJ/kg K) inside a 95 g aluminum (c_{Al} = 0.91 kJ/kg K) calorimeter cup, which are in equilibrium with each other at room temperature, 20.0°C. After the system comes to equilibrium with the sample, the temperature is 22.93°C.

(a) What is the specific heat of the sample? (*Ans*: 0.4398 kJ/kg K)

(b) Looking at this website, what is the identity of the unknown metal? Can you determine it uniquely?

7. It's final's week and you need to cram for you physics final tomorrow. Around 2:00 am, you start feeling pretty drowsy and decide to make some coffee. The freshly made coffee is at a temperature of 94.6°C, way too hot to drink, and you need that caffeine in your system as soon as possible. You decide to pull out some ice cubes to drop in and bring the temperature down. You have 355 ml of coffee in a stryofoam cup.

   (a) The ice cubes start at a temperature of -10.0°C, and are cubes that measure 1 inch (2.54 cm) on each side. How much heat is needed to completely melt one ice cube? Note $\rho_{\text{ice}} = 917\text{kg/m}^3$, $L_{\text{F,ice}} = 79.7$ cal/g, $c_{\text{ice}} = 0.5$ cal/g·C°. (*Ans*: 1272.78 cal)

   (b) If you want to cool your coffee to around 55°C so that you can drink it very quickly, how many ice cubes will you need? Since styrofoam is an effective insulator, we can ignore the effects of the cup. Assume that the coffee has the same specific heat as water. (*Ans*: About 7 cubes)

8. You are trying determine the latent heat of fusion for tin. In your experiment, you heat up 50 g of tin ($c_{\text{Sn}} = 0.0502$ cal/g·C°) until it is just melted, i.e. to a temperature of 232°C. You then carefully pour the molten tin into 500 g of water ($c_{\text{H}_2\text{O}} = 1.0$ cal/g·C°) in a 250 g aluminum ($c_{\text{Al}} = 0.217$ cal/g·C°) calorimeter cup that both start at 20.0°. The final temperature of the system is 22.22°C. What is the latent heat of fusion for tin? (*Ans*: 14.08 cal/g = 58.93 kJ/kg)

9. You have exactly 1 mole of an ideal gas in a piston. You compress it to half its initial volume isothermally at room temperature (293 K).

   (a) How much work do you do on the gas? (*Ans*: -1688.5 J)

   (b) How much heat is released? (*Ans*: -1688.5 J)

10. **Numerical:** In the previous assignment, you randomly generated the velocities for 10,000 nitrogen molecules ($m_{\text{N}_2} = 4.6 \times 10^{-26}$ kg) and then found their root mean square speed, their average speed, and the temperatures that you would assign to that collection of particles based on those speeds, assuming they follow a Maxwell distribution. All of that work was based on looking at the quantities that represent the average energy per particle in some way. We have now learned about the quantity called *internal energy* which should be the total energy of all the particles of a substance and should be equal to

$$E_{\text{int}} = \frac{3}{2} N k_{\text{B}} T = \frac{3}{2} nRT. \tag{14.1}$$

    Let's now test if that relation holds with our randomly generated particles.

    (a) Start by creating a spreadsheet to randomly generate the velocities for 10,000 nitrogen molecules at room temperature, $T = 293$ K. You can do this by downloading another copy of the provided spreadsheet from the previous assignment and following the steps described there.

    (b) Add a column to calculate the *kinetic* energy of each molecule, and then find the sum of this column to calculate the internal energy.

    (c) Compute the expected internal energy using equation (14.1). Calculate the percent difference between this expected value and the value you found in the previous part. Do you values agree reasonably well?

(d) Modify the spreadsheet to generate velocities for 20,000 molecules and redo the calculations. Do the values of internal energy agree better or worse now?

15 | Review

1. Consider the following two vectors

$$\vec{\mathbf{A}} = 5.0 \text{ m } @ \ 25°, \quad \text{and} \quad \vec{\mathbf{B}} = 4.0 \text{ m } @ \ 118°$$

where both angles are measured with respect to the $+x$-axis.

   (a) Find the x and y-components of the vectors and write them down using unit vector notation.

   (b) Find the vector sum $\vec{\mathbf{C}} = \vec{\mathbf{A}} + \vec{\mathbf{B}}$ and give the result in unit vector notation.

   (c) Find the magnitude of the vector sum $\vec{\mathbf{C}} = \vec{\mathbf{A}} + \vec{\mathbf{B}}$, and it's angle relative to the $+x$-axis.

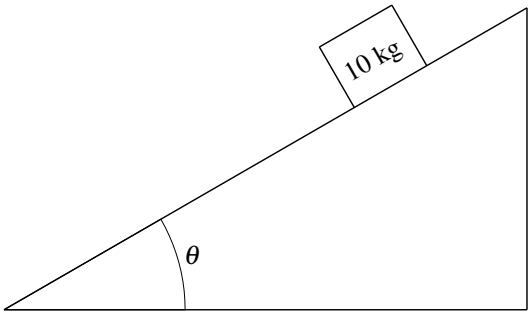
2. An archer is shooting their bow at a target that is $\Delta x = 100$ m away. The bull's eye of the target is at the same height as the arrow as it leaves the bow (i.e. $\Delta y = y - y_0 = 0$). The arrow leaves the bow at a speed of $v_0 = 75$ m/s. Your goal is to determine the angle θ at which the arrow should be fired. Assume that air resistance is negligible.

   (a) Using the horizontal motion of the arrow, derive an expression for the time it will take the arrow to reach the target in terms of Δx, v_0, and θ.

   (b) Using the vertical motion of the arrow, derive an expression for the angle at which the arrow must be fired. **Hint:** You will need the expression from the previous part to replace the time-of-flight of the arrow. Also, note that $2\sin(\theta)\cos(\theta) = \sin(2\theta)$.

   (c) Using your expression and the values given in this problem, find the angle the arrow should be fired at.

3. A block is initially at rest on an inclined plane as shown in the figure below. The angle of the incline is $\theta = 30°$. The coefficient of static friction between the block and the plane is $\mu_s = 0.5$ and the coefficient of kinetic friction is $\mu_k = 0.3$.



(a) Draw and label a free-body diagram for the block.

(b) Using your free-body diagram as a guide, write out the sum of the forces equations for the block assuming the block will remain at rest. Use these equations to find the needed static friction force to keep the block at rest.

(c) Calculate the maximum static friction force the plane is capable of providing. Will the block remain at rest?

(d) Assuming that the block starts moving, either because the static friction is insufficient, or because the block is nudged to get it moving, appropriately modify your sum of forces equations and determine the acceleration of the block.

4. While bored, you are swinging your keys ($m = 200$ g) that are at the end of a 24-inch (60.96 cm) lanyard in a vertical circle.

(a) Draw and label a free-body diagram for the keys at the top and bottom of the circular path.

(b) What is the minimum speed that you can swing the keys so that they continue to move in a circle at the top of the arc? **Hint:** What is the lowest the acceleration can be at the top?

(c) Assuming that you are adjusting the tension in the lanyard so that it continues to move at the same speed as it goes around the circle, what would the tension be at the bottom of the circular path?

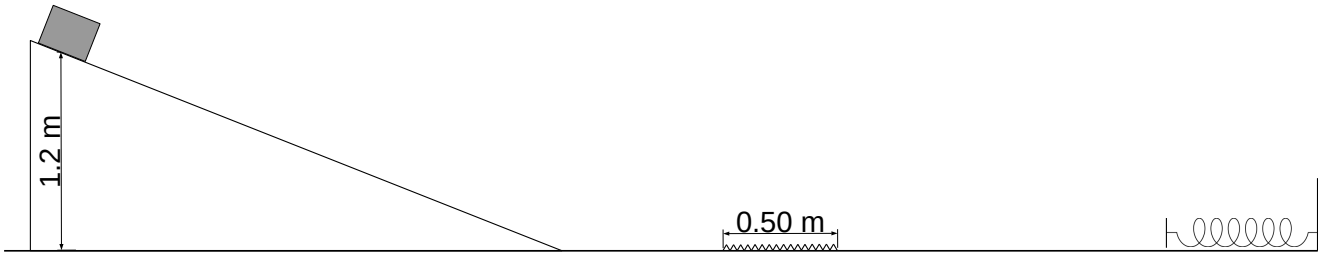
5. After you graduate, you get a job as NASA. One of your first assignments is working with a team that has sent a satellite to orbit and study a large asteroid. One of the reasons for using a satellite to study these asteroids is that we can accurately measure their masses by the speed at which the satellite needs to orbit.

(a) Derive an expression for the mass of the asteroid in terms of the orbital radius, the orbital speed, and Newton's universal gravitational constant, G.

(b) It is found that the satellite needs to maintain 377.8 m/s to maintain an orbit with a radius of 30 km. What is the mass of the asteroid?

(c) What is the period of the satellite's orbit?

6. A block, $m = 2.0$ kg is sitting on a frictionless, inclined plane, initially at rest. The block is 1.2 m, measured vertically, up from a horizontal plane at the bottom of the ramp. The block is allowed to slide down the plane.



(a) What is the speed of the block when it reaches the bottom of the incline?

(b) The horizontal plane has a rough patch, 0.50 m long, which can no longer be considered frictionless and has a coefficient of kinetic friction of $\mu_k = 0.4$. How much work is done by friction as the block slides over this rough patch?

(c) What is the speed of the block after passing over the rough patch?

(d) After the rough patch, the horizontal plane is again smooth enough to be treated as frictionless. The block then runs into a horizontal spring attached to a wall, with spring constant $k = 3500$ N/m. How far will the spring compress?

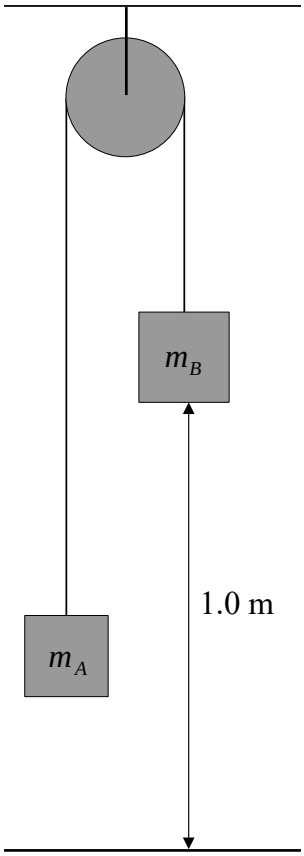
7. A hockey puck (A) is sliding down the ice with negligible friction at a speed of $v_A = 15.0\ m/s$ in what we will call the $+x$-direction. It collides with a second hockey puck (B) of equal mass (160 grams) that is initially at rest. Puck A is seen to move off at an angle of $\theta_A = 27°$ above the $+x$-axis, while Puck B is seen to move off at an angle of $\theta_B = 58°$ below the $+x$-axis.

   (a) What are the speeds, v'_A and v'_B, of the pucks after the collision?

   (b) Is the collision elastic or inelastic? **Note:** You need to have some math to back up your answer here.

   (c) If the pucks are in contact with each other for 2×10^{-3} s, what is the average force that puck A exerts on puck B?

8. Consider and Atwood's machine where $m_A = 1.5$ kg and $m_B = 2.0$ kg and the rope connecting the masses goes over a pulley of $m_p = 0.3$ kg and $R_p = 0.06$ m.



   (a) Find the acceleration of the system if we assume the pulley can be modeled as a uniform disk, $I_{disk} = (1/2)mR^2$.

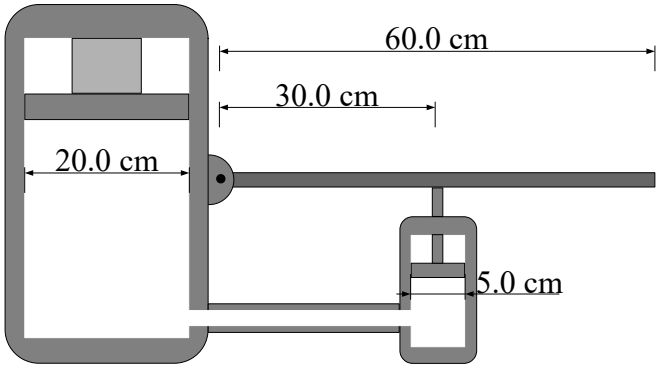
   (b) Find the tensions in the rope on either side of the pulley.

   (c) Find the angular acceleration and net torque on the pulley.

   (d) If mass B starts at rest 1.0 m above the ground, through what angle will the pulley have rotated by the time B hits the ground?

9. Hydraulics make use of Pascal's principle to multiply an input force to a much larger output force. You decide to make a hydraulic press to test the compressive strength of materials. Your input cylinder has a diameter of 5.0 cm, and the output cylinder has a diameter of 20.0 cm. You use a

lever to push down on a piston to increase the pressure in the hydraulic fluid. The lever is 60.0 cm long and the piston is pushed on by the middle of the lever. You prepare samples in such a way that when the lever is horizontal, the output piston is pressed against the sample, which in turn presses against the top of the press.



(a) You place a cube of concrete, 10.0 cm on a side, in the press (compressive strength 20×10^6 N/m^2). How much force will you need to apply to the end of the lever to cause the concrete to break?

(b) You next place a cube of an unknown sample, still 10.0 cm on a side, in the press and find that it takes a force of 1.09×10^4 N on the lever to get the sample to crumble to pieces. What is the compressive strength of this unknown sample?

10. You get an internship with NASA where part of your job is to work with a local middle school to develop an experiment that will be sent with the first astronaut team to walk on the Moon since the Apollo program. You decide to design a simple pendulum to be able to measure the gravitational acceleration on the Moon's surface.

    (a) You want to design the pendulum so that the period is exactly 1.00 s on Earth. How long should you make the pendulum?

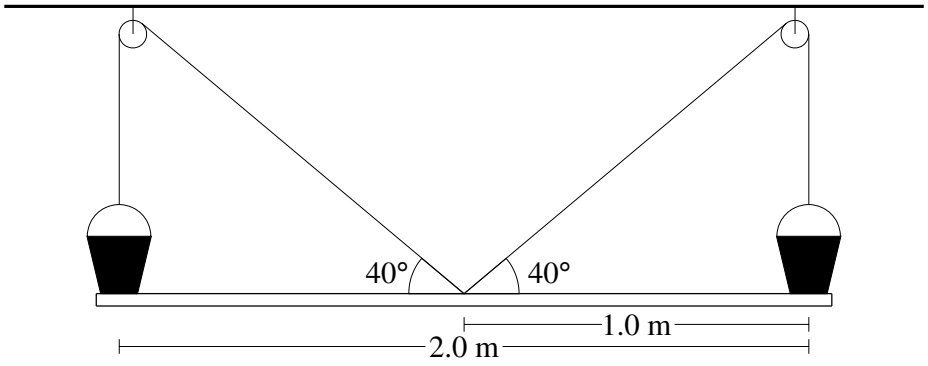
    (b) Once the pendulum arrives on the Moon, the astronauts measure its period and report back to you that it was $T_{\text{Moon}} = 2.42$ s. What is the acceleration due to gravity on the Moon?

11. In lab you are given a small amplitude mechanical oscillator that runs at a fixed frequency of 120 Hz. You run a string over the oscillator on one end and over a pulley on the other. You hang 250 g from the end of the string that goes over the pulley. You start with 1.7 m between the oscillator and pulley, then start sliding the oscillator closer to the pulley. When you get the oscillator to 1.523 m from the pulley you notice a standing wave pattern on the string that has 6 antinodes. (**Hint:** How many antinodes per wavelength?)

    (a) What is the linear mass density of the string?

    (b) You set the distance between the oscillator and pulley to 1.5 m. How much mass should you hang off the string to see a standing wave pattern with 2 antinodes?

12. A uniform plank of mass $M_{\text{p}} = 12$ kg has two ropes connected to its center. These ropes go up to two pulleys at an angle of 40° to the horizontal. They then go down vertically to where they are attached to two buckets of the same mass $m_{\text{b}} = 10$ kg. The buckets also rest against the plank (i.e. there is a normal force, F_{N} between each bucket and the plank). See the diagram below. **Note:** The whole system is completely at rest.



(a) Draw a free-body diagram for one of the buckets and write out the sum of forces on the bucket. Would anything be different for the other bucket?

(b) Draw an extended free-body diagram for the plank. Sum the forces and the torques (about the center) for the plank. What should those equations be equal to? Which one is the only one that matters?

(c) What is the tension in the string?

(d) What is the normal force between the plank and a bucket?